MILITARY SPECIFICATION

GENERATING PLANT, OXYGEN-NITROGEN, TRAILER MOUNTED A/M 26U-1

This specification has been approved by the Department of the Air Force and by the Naval Air Systems Command.

1. SCOPE

1.1 This specification covers the requirements for a trailer mounted, air transportable, self-contained air liquefaction and separation plant for the production of liquid oxygen and liquid nitrogen, designated A/M 26U-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

O-T-634	Trichloroethylene, Technical
P-C-436	Cleaning Compound, Alkali, Boiling Vat (Soak) or Hydrosteam
P-C-437	Cleaning Compound, High Pressure, (Steam) Cleaner
BB-N-411	Nitrogen, Technical
CC-M-636	Motors, Fractional Horsepower (Alternating-Current)
CC-M-641	Motor, Alternating Current (Integral Horsepower, 200 HP and Smaller)
TT-E-489	Enamel, Alkyd, Gloss for Exterior and Interior Surfaces
PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging

Military

MIL-P-116	Preservation, Methods of
MIL-V-173	Varnish, Moisture and Fungus Resistant, for the Treatment of Communications, Electronic and Associated Electrical Equipment
MIL-P-514	Plate, Identification, Instruction and Marking, Blank
MIL-R-3065	Rubber, Fabricated Parts

FSC 3655

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MIL-C-5015	Connectors, Electric, "AN" Type
MIL-T-5021	Tests, Aircraft and Missile Welding Operators, Qualifications
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-I-61366	Inspection, Penetrant Method of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-C-8638	Cleaning Compound, Oxygen Syst-
MIL-D-16791	Detergent, General Purpose Liquid
MIL-C-20696	Cloth, Coated, Nylon, Waterproof
MIL-O-27210	oxygen, Aviator's Breathing, Liquid and Gas
MIL-T-27602	Trichloroethylene, Oxygen Propellant Compatible
MIL-T-27730	Tape, Antiseize, Tetrafluorethylene, with Dispenser

sTANDARDS

Federal

Federal Standard No. 595 Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Markting of US Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-171	Finishing of Metal and Wood Surfaces
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-470	Maintainability Program Requirements, for Systems and Equipments
MIL-STD-471	Maintainability Demonstration
MIL-STD-810	Environmental Test Methods
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods

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MS18045	Battery, Storage, Aircraft, 24-Volt, General Purpose
MS25182	Connector, Plug, Electric Quick Disconnect, Battery
MS33586	Metals, Definition of Dissimilar
AN2552	Receptacle - External Power, 28 Volt DC

DRAWINGS

Air Force

48B 7796	Ring Assembly -- Tiedown, 10,000#
49B 6564	Ring, Eyebolt, 25,000 Pound Carton Tiedown

Navy

3200300	Oxygen/Nitrogen Generating Plant Assembly

PUBLICATIONS

U. S. Air Force Specification Bulletin

Bulletin No. 518 Cargo, Aircraft Compartment Dimensional Data

Navy

NAVAIR 01-85WBB-1-1 NATOPS Flight Manual Cargo Loading
(Navy Model C-2A Aircraft)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society of Mechanical Engineers

ASME Boiler and Pressure Vessel Code
ASME Section VIII Unfired Pressure Vessels

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, New York 10017.)

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National Bureau of Standards

Handbook H2B Screw-Threads Standards for Federal Services

(Application for copies of the above publication should be addressed to the Superintendent of Documents, Government Printing office, Washington, D. C. 20402.)

United States of America Standards Institute

C1-1965	National Electrical Code
S1.4-1961	General Purpose Sound Level Meters, Specification for
S1.6-1967	Referred Frequencies and Band Numbers for Acoustical Measurements
Z24.10	Octane Bend Filter Set for Analysis of Noise and Other Sounds

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street) New York, New York 10016.)

Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies of the above publication should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Preproduction. - This specification makes provisions for preproduction inspection (see 4.4).

3.2 Components. - The plant shall consist of the following major components:

Trailer van assembly
Air source assembly
Air liquefaction and separation assembly

3.2.1 General description. - The major components listed in 3.2 shall constitute a liquid oxygen-liquid nitrogen generating plant which shall be a compact, completely self-contained trailer mounted unit requiring only a supply of aviation fuel and ambient air, plus the necessary absorbents and lubricants to liquefy and separate air into oxygen and nitrogen, The plant shall be mounted on a two-axle, four-wheel trailer, and shall be air transportable. All electrical switch gear, wiring, and lighting equipment, piping, ducting, controls, instruments, and accessory components needed for operation and maintenance of the generating plant shall be provided.

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3.2.2 Component schematic. - The trailer van is shown on figure 1. The simplified flow diagram of the air source and the air liquefaction and separation assembly shown on figure 2 is optional.

3.3 Selection of specifications and standards. - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Parts and materials. -

3.4.1 Standard parts. - In the selection of parts and materials, AN, MS, or MIL standard parts and materials shall be given first consideration. Every practical effort shall be made to use standard, readily available components, and to make the design such that selection is not required when replacement of' components is necessary.

3.4.2 Commercial parts. - Commercial parts having suitable properties may be used where there are no suitable standard parts available. In any case, commercial utility parts such as screws, nuts, bolts, and cotter pins may be used provided:

(a) They are interchangeable with corresponding AN or MS parts without alteration
(b) The corresponding standard part numbers are referenced to the extent practicable on the contractors drawings and parts lists.

3.4.3 Component selection. - Standard, heavy duty, commercially proven components shall be used to the maximum practical extent.

3.4.3.2 Safety devices. - All safety devices shall be of corrosion-resistant commnent materials not adversely affected by extended contact with oxygen, moist air, water, or cleaning solvents such as trichlorethylene.

3.4.4 Metals. - Wherever practicable, lightweight metals shall be used in the construction of the generating plant. All metallic surfaces of the plant that contact high purity liquid oxygen or liquid nitrogen, or that are in a location where they will normally be chilled to an extent causing condensation of moisture from the atmosphere, shall be fabricated from corrosion-resistant materials that do not require paint or other organic chemical. coatings for protection, All other metals shall be either corrosion resistant or treated to resist corrosion resulting from enviromnental conditions specified herein.

3.4.5 Dissimilar metals. - Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

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3.4.6 Sheet metal parts. - Sheet metal parts shall be of sufficient thickness, rigidity, and strength to withstand any conditions which might be normally encountered during transport, field positioning, leveling, service, and maintenance.

3.4.7 Insulation. - Thermal insulation used for the generating plant shall be of the type that is not flammable and that is suitably protected against moisture absorption. The insulation shall be of such a texture that it can be readily removed. and replaced by maintenance personnel in the field.

3.4.8 Gasketing and insulating materials. - Plastic, rubber-like, rubber, or similar gasketing and insulating materials shall be compounded to insure their suitability for the intended application, and, where applicable, their resistance to hydrocarbons or low temperatures. Where polytetrafluoroethylene gasketing is used, this material shall be impregnated.tith not less than 25 percent pulverized glass fiber filler to improve its dimensional stability under compression. Packing, gaskets, seals, and other nonmetallic components that come into contact with high purity oxygen shall be compounded of materials compatible with use in oxygen atmosphere. They shall be suitable for use under all conditions to which they will be subjected and shall not adversely be affected or deteriorated by continuous use in an oxygen or nitrogen atmospherE,

3.4.9 Rubber materials. - Rubber for components such as hoses, seals, and insulators shall be compounded in accordance with MIL-R-3065.

3.4.10 Combustible materials. - Readily combustible materials shall not be used.

3.4.11 Material safety. - Materials shall not be used if the use of that material under the conditions specified herein will cause an evolution of toxic product detrimental to the health of the operating personnel or a product which may be classified as corrosive, combustible, or explosive in an oxygen-enriched environment.

3.4.12 Low temperature service. - Materials subjected to cryogenic temperatures (-361° F) shall have sufficient ductility and impact strength for the senice~ Carbon steels shall not be used at cryogenic temperatuers.

3.4.13 Pine thread lubricant or sealant. - Tetrafluoroethylene tape conforming to MIL-T-27730 shall be applied to threads prior to assembly of all pipe fittings subject to contact by liquid or gaseous oxygen or nitrogen. The tape shall be applied starting with the third thread to prevent contamination of the system.

3.4.14 Fungus-Proof materials. - Wherever practical, materials that are nutrients for fungi shall not be used. Where used and not hermetically sealed, they shall be treated with varnish conforming to MIL-V-173. However, if they are used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.5 Design. -

3.5.1 General design. - The generating plant shall be a compact, lightweight, unit complete in design for:

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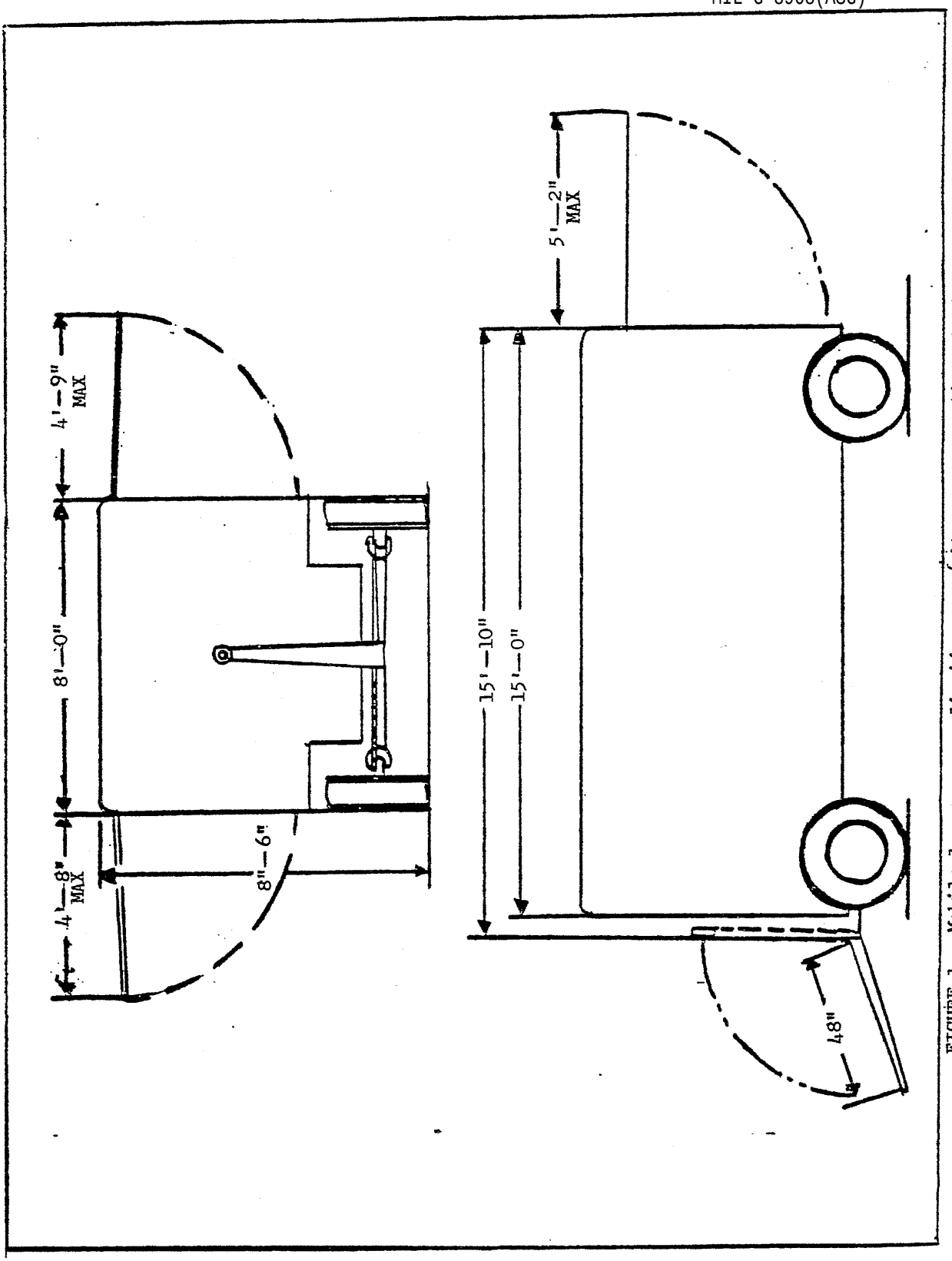


FIGURE 1. Mobile low pressure liquid oxygen/nitrogen generating plant

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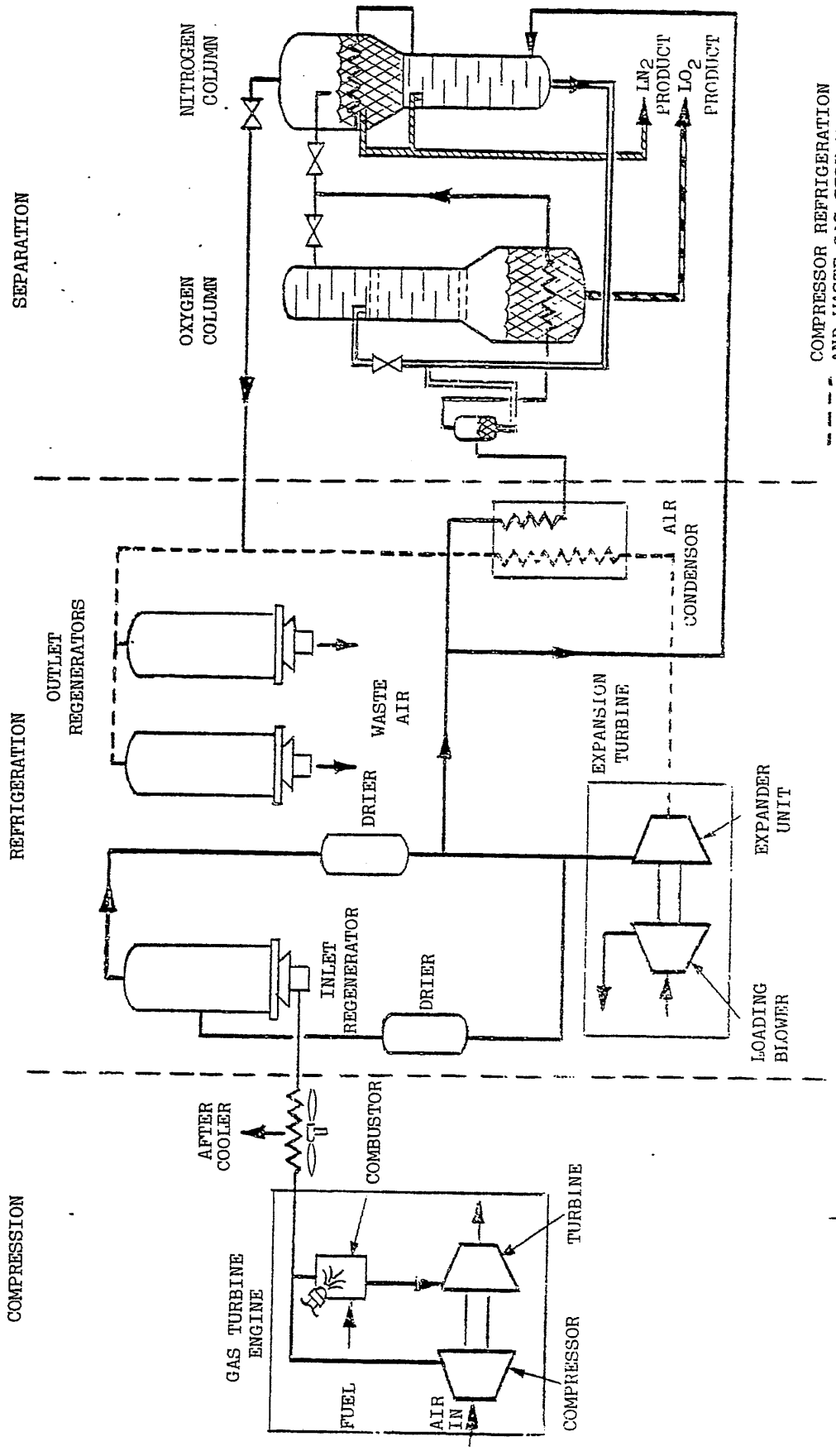


FIGURE 2. Flow schematic (simplified)

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(a) Air, surface, or water transportation to an advance or remote tactical air base under emergency conditions
(b) Being quickly set up and put into service to generate and deliver high purity liquid oxygen and nitrogen on any reasonably level ground in the vicinity of the air base without prior site preparation other than a removal of rock and foilage and, if required, the installation of M-2 plating or other temporary construction material to serve as level foundation or 'pad where ground texture will net properly support the weight of the plant during periods of adverse weather conditions
(c) Delivering liquid oxygen or liquid nitrogen through transfer hoses into receiving tanks requiring a maximum of 5 psi for transfer and fill
(d) Protecting the generating plant and the operating personnel under the environmental conditions specified herein
(e) Being quickly shut down and readied for movement to a new tactical location
(f) Being operated, maintained, and repaired military personnel at a level commensurate to their training and provisioning, under combat conditions at an advance tactical air base.

3.5.2 Configuration. - The generating plant overall configuration shall be that of a fully enclosed, van-type, four-wheeled trailer (see figure 1) designed to meet the requirements of MIL-M-8090 as specified in type III, group C mobility. All primary instrumentation and primary controls required for normal operation shall be located on one operating side of the trailer and shall be designed for operation from the ground level. Access to the operator's control panel and to the other components requiring service or maintenance shall be provided through readily open and closed weather-protective doors or panels.

3.5.3 Reliability. - The generating plant shall have a mean-time-between-failures (MTBF) of not less than 240 hours of continuous, full capacity, production at not less than 0.60 confidence. A failure is defined for the present purpose as any malfunction or breakdown that could prevent satisfactory performance within the' limits specified herein or that could prevent safe operation for the primary functions of producting and delivering at rated capacity liquid oxygen and liquid nitrogen meeting the requirements specified herein. Malfunctions that do not interfere with performing the required functions of producing and delivering at rated capacity liquid oxygen and liquid nitrogen meeting the requirements specified herein, such as a loose bolt in a non-critical application, need not be counted as reliability failures,

3.5.4 Construction. - The generating plant shall be constructed to withstand the strains, shocks, vibrations, and other detrimental conditions incident to operation, maintenance, shipping, and storage with minimum loss of service the for maintenance, repair, and periodic servicing. Provisions shall be incorporated to insure protection for the equipment from damage that might occur as a result of temperature variations and stresses occurring during operation and transport.

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3.5.4.1 Component mounting. - Components that need not be removed or repositioned for use shall be securely mounted to the generating plant in a manner that insures against damage or unnecessary movement during operation and transport, All sub-bases and other fixtures needed for safe and secure component mounting shall be provided and installed.

3.5.4.2 Tubing, lines, and electrical wiring. - All tubing, lines, and electrical wiring shall be located in protected positions, securely fastened to framework or body structures, and provided with protection sleeves, grommets, or similar devices at each point where they pass through members, except where a feed-through connector is provided, "

3.5.4.3 Trailer van body. - The trailer body, roof, doors, and panels in the closed position shall provide adequate protection for the plant from damage during storage or transport under weather conditions specified in 3.6.2 and shall be so designed as not to warp, shrink, or delaminate in doing so. The doors and panels shall be equipped with tiedown provisions for securing in the open position to withstand gusts of wind. The roof, doors, and panels in the open position shall support, without permanent deformation, a load of 40 psf.

3.5.4.4 Center of gravity. - The generating plant overall center-of-gravity shall be as low as practicable, Components and parts on the generating plant shall be so mounted and positioned that the resultant forces (as calculated from the overall center-of-gravity) will be to the rear of the front axle when the generating plant is decelerated at a rate of 18 feet per second per second.

3.5.5 Maintainability. - Parts and assemblies shall be located and mounted to provide adequate clearance for repair and other maintenance and where practicable, to permit removal and replacement of any part or assembly by removing or disconnecting only mounting bolts or fasteners, panel doors, tubing, or control cables to the part or assembly being removed. The generating plant shall be designed and constructed in accordance with the design guidelines in MIL-STD-470, paragraph entitled, "Establish Maintainability Design Criteria", except rather than merely considered> the guidelines shall be applied wherever appropriate to the subject class or type of equipment and wherever they do not conflict with other requirements specified herein. Where practicable, the design and construction of the generating plant shall permit maintenance and servicing under adverse weather extremes. All maintenance, servicing, and adjustment which may be economically practicable at lower then depot level shall be capable of being performed using only common handtools and maintenance equipment. *

3.5.5.1 Operating clearances. - To the maximum practicable extent, maintainability provisions shall incorporate features insuring operating clearance for facilitating maintenance and servicing at low ambient temperatures by personnel wearing heavy gloves or mittens and handicapped by bulky clothing and footwear.

3.5.5.2 Overtorque. - Intricate locking devices, controls, and threaded fasteners that can be easily overtorqued by personnel lacking feel through thick gloves or mittens shall be avoided where possible.

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3.5.5.3 Mountings. - Where improper installation of an item could cause malfunctioning to that item or the system in which it is installed, an unsymmetrical mounting means shall be provided. This mounting shall be so designed that the item can only be installed in its proper operating positions If an Unsymmetrical mounting means is not practical, the items shall be so mounted that its proper operating position can readily and visibly be determined by service maintenance personnel.

3.5.5.4 Covers and plate fasteners, - Covers and plates that must be removed for frequent component adjustment or component or part removal shall be equipped with substantial quick-disconnect fasteners

3.5.5.5 Components. - All components having moving parts which are probable sources of maintenance, such as valves, pumps, and fans, shall be located in such a fashion that they can be removed for maintenance or replacement.

3.5.5.6 Fittings. - Wherever practicable, the plant shall be designed to include AN and MS standard disconnect plugs, receptacles, and multiple line connectors conforming to MIL-C-5015 in the electrical systems, and readily detachable and attachable-type fittings in the lines to permit rapid component removal and replacement. Disconnect points shall be clearly indicated and identified,

3.5.5.7 Lubrication, - Wherever practicable, lubricated-for-life components or components not requiring lubrication shall be used in the generating plant construction and design. Grease and oil seals shall be located and installed to provide maximum accessibility for inspection and replacement, and constructed to minimize damage from excessive lubricant quantity or application pressures.

3.5.5.7.1 Parts subject to wear. - The generating plant shall be designed and constructed to provide for proper lubrication of parts subject to wear, such parts shall be constructed, located, and installed to be readily renewable. All grease fittings and other lubrication points and lubricant drains shall be readily accessible,

3.5.5.8 Protective covers. - All cover plates, gaskets, and fittings necessary for protection of contained apparatus during operation, storage, and transport of the generating plant shall be provided. Covers shall also be provided to protect air, oil, fuel and product lines when such lines are opened because of disuse or from component removal.

3.5.5.9 Wiring. - All electrical wiring shall be numerically coded except trailer lighting wiring which shall be color coded in accordance with SAE standards. Identifying numbers shall be printed not more than 3 inches apart directly on the wire insulation. Wiring runs shall be located to enable the accessibility for maintenance and checkout.

3.5.5.10 Equipment location. - All components having moving parts which are probable sources of maintenance, shall be installed in a manner to permit readily removal without removing the basic cold box insulation for maintenance or replacement.

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3.5.5.11 Interchangeability. - All parts having the same manufacturer's part number' shall be directly and completely interchangeable with respect to installation and performance, The generating-plant shall be designed to provide for the maximum practicable interchangeability of hardware and fastening devices by using the minimum number of types and sizes of bolts, screws, nuts, washers, and other common hardware.

3.5.5.12 Maintenance downtime. - The generating plant shall be designed and constructed to petit maintenance by military maintenance personnel in not more than the downtimes specified in 3.5.5.12.1 and 3.5.5.12.2. No more than two technicians shall be required simultaneously for any one task.

3.5.5.12.1 Corrective maintenance downtime. - The mean corrective maintenance downtime for all levels of maintenance below depot level shall be not more than 2 hours with a related maximummafi.tens.nce downtime of not more than 4 hours.

3.5.5.12.2 Maximum preventive maintenance downtime. - Maximum maintenance downtime, excluding depot maintenance and major overhaul, shall not be more than 4 hours at the 0.9 percentile point.

3.5.6 Operational life. - The overall equipment design shall be such as to minimize loss of production time for servicing, impactions, maintenance, repairs, and similar operations.

3.5.7 Safety. - The construction of this equipment shall provide for maximum safety to personnel during operation installation, or interchange of assemblies and component parts. All surfaces that will attain temperatures likely to be harmful to personnel shall be rendered safe either by design location or by insulation. Surfaces that are exposed and are at temperatures sufficient to condense ambient air shall be insulated. The necessary silencing equipment shall be included to permit continuous 12-hour operation without detrimental effects to the health of an operator. The acoustic noise levels produced by the generating plant under any of the conditions specified herein shall not-exceed values specified in 4.6.11.
generating plant, and the product transfer hoses shall be equipped with safety features. Safety devices which are common in sound design practice for equipment of this type shall be used.

3.5.7.1 Compartment lighting. - Where components which require servicing or frequent maintenance have been installed in compartments, adequate lighting shall be installed within the compartment to provide the visibility necessary for safety and accurate maintenance.

3.5.7.2 Pressure vessels. - Unless otherwise specified herein, all pressure vessels, lines, connections, and components of the generating plant that operate under pressure shall be designed and fabricated in accordance with section-VIII of the ASME Boiler and Pressure Vessel Code.

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3.5.7.3 Pressure relief devices. - Pressure relief devices shall be provided in any line or component that can be so isolated by closing valves, dead-ending, or otherwise, that dangerous pressures could develop. Relief valves shall be provided for normal relief of pressure. All pressure relief devices shall be installed in readily accessible locations, Provisions shall be incorporated, however, to insure that the discharge of pressure may be accomplished without (a) damage to equipment or hazard to personnel and (b) creating fog where it will obscure controls, instruments of fittings requiring the attention of the operator or maintenance personnel. Safety valves shall be sized in accordance with the ASME Unfired Pressure Vessel Code and selected from the list published in the "Safety Valves and Relief Valves approved by the National Board.

3.5.7.4 Blow out-type safety heads. - Blow out-type safety heads may be provided where pressure will not normally have to be relieved, but where protection is necessary, such as for the air processing and separation insulated casings.

3.5.7.5 Guards. - Moving parts such as belts, chains, gears, linkages, and connecting rods shall be provided with guards which will prevent accidental contact by personnel. The guards shall be constructed to permit access for inspection of enclosed equipment, and to permit ready removal when necessary for maintenance or repair.

3.5.7.6 Sharp edges. - Sharp edges, projections, and hinged devices with hazardous characteristics shall be avoided.

3.5.7.7 Safety devices. - Safety devices shall be installed on the air source assembly, automatic shutdown or visible, to preclude damage to the engine if abnormal operating conditions exceeding those specified by the manufacturer occur. An interlock shall also be provided in the engine starting circuit to prevent initiation of start sequence until air inlet door is secured in open position.

3.5.7.8 Cleaning solvent drains. - Provisions shall be included to insure that cleaning solvents and flushing liquids can be readily and thoroughly removed from the columns, heat exchangers, associated components, and from the piping and connections following cleaning operations. The use of readily removable and replaceable threaded plugs or similar devices in locations where it would be impractical to provide drain valves will be acceptable.

3.5.7.9 Noise suppression. - Noise attenuation, suppression, and silencing equipment and provisions shall be included to the maximum practicable extent to permit continuous operator attendance without detrimental effects on personnel health.

3.5.7.10 Lacking devices. - All screws, pins, and bolts shall be equipped with locking devices. Safety wire, self-locking nuts, cotter pins, and lockwashers will be acceptable, Where practicable, lockwashers shall be secured-to bolts or screws. Safety wire and cotter pins shall be fabricated from corrosion-resistant steel or other corrosion-resistant material.

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3.5.8 Certification of welders. - All welding shall be accomplished by welders certified in accordance with class A of MIL-T-5021.

3.5.9 Transportability - The generating plant shall meet the air transportability requirements of MIL-A-8421. Additionally, when specified (see 6.2) the plant shall be compatible with amphibious type operations. Consideration shall be-given to embarking and debarking the plaint in amphibious ships of the IST type, The plant shall be capable of being moved through salt water to the depth equal to the height of the wheel without detrimental effect to equipment.

3.5.9.1 Mobility. - The generating plant vehicle assembly shall be of full trailer configuration designed to meet mobility requirements of type III, group c of MIL-M-8090.

3.5.9.2 Aircraft loading and transport. - The generating plant design and configuration shall be such as to provide for loading and transport by type C-130 and larger aircraft, Applicable aircraft cargo compartment and loading door dimensional data are contained in Bulletin No. 518 and NAVAIR 01-85WBB-1-1, for Navy C-2A aircraft,

3.5.9.3 Transport acceleration forces. - The generating plant shall be designed and constructed to withstand the flight and taxiing acceleration forces specified in MIL-A-8421 during transport and handling. The plant shall withstand like forces as . imposed in MIL-A-8421 while being transported on railroad car, seagoing vessel, or highway transporter.

3.5.9.3.1 Vibration forces, - The generating plant shall not be damaged by the transport vibration forces specified under Method 514, Procedure X of MIL-STD-810 for category g equipment, when completely assembled and ready for shipment.

3.5.9.4 Emergency landing acceleration forces, - The completely assembled generating plant prepared for transport shall withstand the emergency landing load shock forces specified in MIL-A-8421, except that the shock pulse shall be an approximate one-half sine wave of not less than 30 milliseconds or more than 40 milliseconds' duration without breaking loose from its tiedowns or parts of the equipment leaving the original envelope of the vehicle,

3.5.9.5 Amphibious operation. - When specified (see 6.2), the generating plant design, configuration, and construction shall provide for loading into, transport by, and disembarking through the surf and over the beach from amphibious ships of the IST type. The design configuration, and construction shall also be such to permit fording or being moved through fresh or salt water to a depth equal to the height of the wheels and tires without damage to or detrimental effect on the equipment.

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3.5.9.6 Hoisting and tiedown provisions. -

3.5.9.6.1 Lifting rings. - The generating plant shall be provided with lifting rings located at accessible points that will permit cable attachment and hoisting without damage to the trailer van assembly or other generating plant components. The lifting rings shall permit lifting of the complete generating plant when prepared for shipment, considering vertical accelerations of 3g. The lifting rings shall be so located on the generating plant that transportation personnel can readily rig safe lifting slings from common cable and spreader bar components for lifting by a single hook, overhead crane.

3.5.9.6.2 Lifting eyes. - Lifting eyes or other hoisting attachments shall be so provided on heavy components tnat they may be readily removed from or replaced on the trailer with overhead hoisting facilities. Insofar as it is practicable, all individual components shall be located and installed to permit their removal without removing other components.

3.5.9.6.3 Tiedown provisions. - The generating plant trailer body frame structure shall be provided with sockets for attachment of tiedown rings confirming to drawing 4936564 and drawing 48B7796 to insure safe and secure tiedown for transport by railroad, aircraft, highway carrier, or ship when it is subjected to the acceleration forces specified in 3.5.9.3 and 3.5.9.4 The number and location of sockets shall be such as to insure tiedown in all directions (both sides, front, and rear) and in any case shall include one near each corner of the body. A tiedown ring conforming to drawing 48B7796 shall be installed in each socket.

3.5.9.7 Towing tractor compatibility. - The generating plant shall be designed to permit use with, and prevent damage to the running gear and other components from use with, towing tractors having pintle hooks from 10 through 36 inches above ground level and requiring up to 48 inches sideways from the pintle hook to clear the tractor wheels during short turns or backing.

3.5.9.8 Pushing provisions. - The generating plant shall be designed to include provisions for permitting use of a vehicle behind the generating plant to push from the rear in assisting the traversing of difficult terrain, deep mud, or sand.

3.5.10 Winterization provisions. - The generating plant shall be designed to operate without the use of external or special winterization provisions in any ambient temperature between -30° F and 125°F. The design shall provide for detachable weather shelter and heating when necessary for operating personnel working under these temperature; conditions. The design shall also permit and provide for use of external heaters as specified by the procuring activity for starting and operating if necessary in ambient temperature down to and including -65°F.

3.5.10.1 Oil dilution. - Oil dilution shall not be an acceptable winterization provision.

3.5.11 Supplies. - The generating plant shall not require operating supplies other than fuel, atmospheric air, lubricants, chemical and calibration gas for product analysis and spare parts for maintenance and repair, and the necessary tools.

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3.5.11,1 Fuel supplies. - The generating plant shall be designed to perform as specified herein when supplied with JP-4 fuel or JP-5 fuel conforming to MIL-T-5624.

3.5.11.2 Lubricants. - The generating plant shall be designed to perform as specified herein when components needing lubrication are lubricated in accordance with MIL-STD-838.

3.5.12 Screw threads. - Screw threads shall conform to Handbook H28.

3.6 Operation. - The generating plant shall meet the operating requirements . specified herein when tested in accordance with inspection methods specified in 4.6,

3.6.1 Operating cycle. - Tne plant shall be capable of continuous operation for a 240-hour production run without deriming, except for servicing as provided in 3.6.30

3.6.2 Environmental conditions, - The generating plant shall operate under and withstand the following environmental conditions without damage or malfunctioning.

3.6.2.1 Operating temperature. - The generating plant shall perform all functions required herein in any ambient temperature between -30°F and 125°F, including exposure to solar or equivalent heat radiation of 100 to 120 watts per square foot at the higher temperature, without shelter or other than self-contained winterization provisions.

3.6.2.2 Storage temperature. - The generating plant shall not be damaged by storage, in any ambient temperature from -65°F to 165° F.

3.6.2.3 Humidity. - The generating plant shall not be damaged by storage or operation in relative humidity conditions up to and including 100 percent at the temperatures specified in 3.6.2.1, including conditions wherein condensation takes place in the form of water or frost.

3.6.2.4 Fungus.- The generating plant shall not be damaged by exposure to moist fungus growth such as encountered in tropical or sub-tropical climates.

3.6.2.5 Salt atmosphere. - The generating plant shall not be damaged by operation or storage in an atmosphere containing salt-laden moisture, such as encountered near bodies of salt water and in transportation on shipboard.

3.6.2.6 Salt water immersion. - The generating plant shall not be damaged by immersion in salt water to a depth equal to the height of the tires, as could be expected to occur during fording or when being moved through salt water during an amphibious operation.

3.6.2.7 Rain. - The generating plant shall not be damaged or deteriorated by operation or storage under heavy rainfall such as encountered in any locale, including the tropics. When the housing assembly is closed for shipment or storage, water from rain shall not penetrate to the interior of the housing.

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3.6.2.8 Sand and dust. - The generating plant shall not be damaged by operation or storage in an atmosphere containing airborne sand and dust particles such as encountered during operation in or near a desert area.

3.6.2.9 Wind. - The generating plant shall not be damaged by storage or operation in wind velocities up to and including 70 tiles per hour, with wind gusts up to and including 125 miles per hour.

3,6.2.10 Snowload. - The generating plant shall not be damaged or any shelter component collapsed by a snowload of 40 psf both when closed for transport or storage and when configured for operation in cold climates.

3.6.3 Endurance. - The generating plant shall be capable of not less than 240-hour continuous liquid production without necessity for a shut down, thaw, or deriming, During such a period, a shut down of not more than 30 minutes during each 8-hour period and not to exceed 90 minutes in any 24-hour period will be permitted for servicing, adjustments, inspections, minor repairs, and preventive maintenance if such shut downs do not reduce the plant production average below that specified under 3.7, Reactivation of adsorbent beds and similar procedures normally performed while the equipment is operating and delivering product are not affected by this requirement as long as they do not necessitate shut down, interference with normal operational procedures, or reduction of the average product rated specified in 3.7.1.

3.6.4 Thawing. - upon completion of a 240-hour continuous production period, the plant shall be completely thawed, serviced, and inspected. It shall be ready for the next production run within 12 hours.

3.7 Performance. - The generating plant shall meet the performance requirements specified herein when tested in accordance with the test methods specified in 4.6.7.1, 4.6.7.2, and 4.6.7.3.

3.7.1 Rating. - The nominal rating of the plant shall be defined as the product output at the purity specified in 3.7.4.1 and 3.7.4.2, at sea level conditions (760 mm barometric pressure) and 70°F ambient temperature and a relative humidity of not less than 75 percent. At this condition, the product output shall not be less than 2.0 tons per 24-hour day of either oxygen or nitrogen, The plant shall be capable of producing not less than 1.6 ton per day of liquid oxygen when operating the generating plant (ambient) and with the intake air at a temperature of 122° F and a relative humidity of 50 percent.

3.7.2 Starting time. When all the generating plant components are initially at or above the temperature specified in 3.6.2.1, the plant shall cool down and achieve the production rate specified herein within 8 hours after starting the’air source assembly.

3.7.3 Restarting time. - During a 240-hour production period, the plant shall resume rated production within an additional 30 minutes or an additional time equal to the length-of the shut down period whichever is less.

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3.7.4 Product quality. -

3.7.4.1 Liquid oxygen. Liquid oxygen product shall conform to type II of MIL-O-27210.

3.7.4.2 Liquid nitrogen. Liquid nitrogen product shall conform to type II, class 1, grade B of BB-N-411.

3.7.5 Product pressure. -

3.7.5.1 Liquid oxygen. - The liquid oxygen product shall be discharged at a pressure of not less than 5 psig.

3.7.5.2 Liquid nitrogen. - The liquid nitrogen product shall be discharged at a pressure of not less than 30 psig.

3.7.6 Acoustic noise level. - The acoustic noise levels produced by the generating plant during operation under any of the conditions specified herein shall not exceed the values specified in table I when measured as specified in 4.6.11.

TABLE I. Acoustic noise level limits

Octave band center frequency (Hz)	Octave band pressure level in decibels (Re 0.0002 Dynes/Cm^2)
63	98
125	91
250	86
500	83
1000	81
2000	80
4000	80
8000	80

3.7.7 Electromagnetic interference limits. - The generating plant shall not emit electromagnetic interference radiation in excess of the limits established for RE05, class IIIC items in MIL-STD-461.

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3.8 Details of components.

3.8.1 Trailer van assembly. - The trailer van assembly of the generating plant shall be designed to meet air transportability requirements of MIL-A-8421 and type III; group C mobility requirements of MIL-M-8090. The running gear shall be a four-wheel type; front wheel steerable, with leaf spring and shock absorbers, The trailer van assembly shall utilize 8.25 x 15 tires and wheels and shall be equipped with running lights, air-over-hydraulic four-wheel service brakes, and mechanical parking brakes. The towing connection shall be a 48-inch draw bar with a lunette eye. Fenders shall be included. Finish and marking shall be as specified by contract and Navy drawing 3280300 and-all drawings listed thereon.

3.8.2 Air source assembly. - The air source assembly shall be a self-contained unit which includes the functional components, as well as an enclosure, associated equipment, and the controls required for operation, Any exhaust discharge shall be directed vertically up on the side opposite from the operator's main control station. Both the air inlet duct and. the dischage vent shall be at least 6 feet above ground level. The inlet duct shall be shielded from foreign object ingestion of 1/4 inch or larger foreign object or accidental blockage.

3.8.2.1 Packaging. - The air supply unit shall be packaged as a separate, removable, independent operating module. The enclosure shall be weather-tight and shall have hinged covers on the instrument and control panels for the air intake and exhaust. The ambient air intake and the exhaust shall be arranged so that intake of exhaust gases shall be held to a minimum. Readily removable access panels shall be provided to facilitate normal maintenance activities.

3.8.2.2 Mounting. - 'Provisions shall 'be included for mounting of the air source package in a manner that permits installation or removal from the mobile assembly by standard fork lifts. Removal shall require only disconnect of the process air and electrical connection, and removal of the holddown bolts or fasteners which secure the air source package to the plant structure.

3.8.2.3 Detached operation. - The air source package shall be capable of full operation when removed from the trailer van assembly with no electrical or fluid connections to the trailer van assembly. During such operation, all gages, meters, controls, and safety features normally used or present for operation shall be operative without changes or modifications to the electrical or Fluid systems, other than closing the bleed air and the emergency shut-off circuits opened by the detachment of the air source package.

3.8.2.4 Accessories, - The air source package shall include all the accessory items such as oil cooler, starting system, lighted instrument panel and other equipment required to make the air source fulfill the operation requirements specified herein.

3.8.2.5 Duty cycle. - The air source shall be suitable for operating 10 days, 22-1/2 hours per day. The unit may require stopping for minor servicing for no longer than 30 minutes at 8-hour or longer periods.

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3.8.2.6 Auxiliary air source. - Connections for connectors as specified by the procuring activity shall be provided to receive and operate on air and electrical power from external sources as an optional mode of operation.

3.8.2.7 Consumable supplies. - The air source shall require no consumable supplies other than fuel, lubricating oil, filter elements, and minor spare parts for extended field operation,

3.9 Air liquefaction and separation assembly. (Optional) Where used, the following shall apply.

3.9.1 Minimum subsystem requirements. - The air liquefaction and separation and liquefaction assembly shall, as a minimum, contain the following major subsystems:

(a) After cooler system
(b) Air purification subsystem
(c) Refrigeration subsystem
(d) Distillation subsystem
(e) Cold box

It shall be possible to visually confirm operation of all rotating components in these compartments. All the low temperature equipment shall be thermally insulated, The insulation shall be protected by readily removable panels which shall be sealed to preclude the entrance of moisture-laden air. A positive purge of dry air or nitrogen shall be maintained on the insulation casing, which shall be provided with a safety relief to prevent buildup of excessive pressure.

3.9.2 Aftercooler subsystem. - The incoming compressed air shall pass through an aftercooler of sufficient heat transfer capacity to maintain a process air temperature less than 25° F above ambient temperature. Downstream of the aftercooler, the air shall pass through a water separator. Provision shall be made for continuous drain of the separated water from the system in a visible location.

3.9.3 Air purification subsystem. - Provisions shall be made to remove water vapor, hydrocarbons, and other contaminants from the process air in the gas phase so as to absolutely prevent dangerous build-up of hydrocarbons. The hydrocarbon removal equipment shall be sized to prevent break-through of hydrocarbons over an operating period of 1-1/2 times that for the required removal equipment operating cycle.

3.9.4 Refrigeration subsystem. - The refrigeration subsystem shall be simple, rugged, and compact. Adjustable nozzles or other complex systems shall not be used in the refrigeration producing component. A lubrication system of adequate capacity shall be provided containing the necessary tooting, pumping, filtering, and controls required for operation, The lubrication system shall be designed to positively prevent intrusion of oil into the process air.

3.9.5 Distillation subsystem. - The distillation subsystem shall contain no more than two distillation columns. No process transfer pumps shall be used in this subsystem or in any other part of the generating plant.

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3.9.6 Cold box. - The number of mechanical joints on the inside of the cold box shall be kept to a minimum. All the auxilliary piping required to safely operate, thaw, and sample shall be permanently installed with shut-off valves on the control panels. Drain lines shall be provided in the bottom of each liquid pot or sump. The unit shall have provisions for thaw of the plant. No moving parts or components with moving parts excepting valve discs or gates, valve stems and the refrigeration producting component shall be contained in the cold box. Piping inside the cold box shall not utilize bellows connections for any purpose, except as required for the refrigeration-producing component.

3.9.7 Product discharge interface. - The liquid nitrogen and liquid oxygen discharge ports shall be located at the rear of the trailer, discharging in the rearward direction, Both products ports shall be located between 3 feet and 6 feet above ground level. The ports shall be located in an enclosed space with clearly marked quick access doors provided for the connection. hose exits. The plant discharge connections for liquid oxygen and liquid nitrogen shall be 1/2-inch jacketed male bayonets, complete-with dust cover assemblies, The liquid product discharge connections shall be sufficiently different that coupling of the oxygen product hose to the nitrogen discharge connection or coupling of the nitrogen product hose to the oxygen product discharge connection will be impossible.

3.9.8 Thaw provisions. - The unit shall have provisions for supplying hot air for thaw, During all operation, the gas/liquid flow direction through all packed beds shall be in the vertically down direction, and, additionally, the beds shall be fitted with means to absolutely prevent bed particle loss. An air vaporizer shall be provided to vaporize the output of the plant, should delivery of the product into the ranks be interrupted. The vaporizer shall also be capable of vaporizing the liquids remaining in the plant on shutdown.

3.9.9 Sampling provisions. - Sampling lines from all significant process points shall be located on the control panel and provided with shutoff valves. The following shall be the minimum requirement for sampling ports:

(a) Oxygen column product
(b) Nitrogen column -product
(c) Oxygen column overhead waste
(d) Nitrogen column overhead waste

3.20 Fuel system. - The air source assembly shall be suitable for operation using MIL-T-5624, grades JP-4 and JP-5 aircraft fuel. The fuel tank shall be lightweight and collapsible, having 1,500 gallons capacity. The fuel tank shall be supplied complete with all fittings, filters, flexible piping and other components required to permit location of the fuel tank at least 25 feet from the air source enclosure. The tank fill connection shall be provided with an adapter as specified by contract (see 6.2).

3.11 Electrical systems. -

3.11.1 Power rating. - The air source package shall include a 120/208-VAC, 400 cycle, 3-phase alternator of sufficient rating to operate all electrical components of the generating plant. All electrical equipment necessary to regulate and control the electrical output shall be included.

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3.11.2 DO supply. - Tine air source package shall include a source of dc voltage for battery charging and control power which shall have sufficient capacity to operate and maintain all equipment requiring nominal 28V dc power, .

3.11.3 Voltage regulation. - Voltage output shall be automatically controlled to maintain a nominal 208 volts (not less than 200 volts and not more than 216 volts) under all load conditions.

3.11.4 Battery, - The plant shall 'be provided with a battery (MS18045-42) which has terminal connections of the quick disconnect type, A mating connector (MS25182-2) shall be used. The battery charging circuit shall be automatically controlled to maintain the battery charge.

3.11.5 External power receptacle, - An external power receptacle conforming to AN2552 shall be provided on or within the air source package at a location convenient for the operator standing on the ground alongside the plant to attach a power cable. The external power receptacle shall be connected into the starting system so that an external power source can be used to supply electrical power to the starting motor and to the emergency lighting system when the battery is disconnected or removed. There shall be a "make before brake" switch in the battery circuit. The external power receptacle shall be equipped with a readily removed cover to protect against accidental contact with or damage to the receptacle pins. If a removable type cover is provided, it shall be securely attached to the receptacle installation by a chain or other means to prevent loss. A caution place warning against connecting a power cable to the external power receptacle unless the battery is removed or disconnected from the circuit shall be mounted adjacent to the receptacle.

3.11.6 Control voltages. - The voltage of electrical power used at all manually actuated switches and pushbutton stations and for all control circuits used in operating the generating plant shall not exceed 120 volts.

3.11.7 Wiring, - AU electrical control wire shall conform to the National Electrical Code C1-1965. Wiring for running gear lights shall conform to type III, group C of MIL-M-8090. Wire shall be numerically coded by identifying numbers printed not more than 3 inches apart directly on the wire insulation. Wiring runs shall be located to enable accessibility for maintenance and checkout,

3.11.8 Lighting. - The plant shall be equipped with all lighting required for night operations. Lighting shall consist of low level instrument panel lighting, lighting within the operating area, a minimum of 4-flood lights for illumination of the general area. Each lighting group shall be controlled separately.

3.11.9 Blackout lighting. - Lighting arrangements suitable for permitting safe operation of the generating plant under blackout conditions without visible light showing at a distance of 250 or more feet from the generating plant shall be incorporated. The nature of the blackout lighting system shall be such that it does not require closing of ventilation and access openings that should be open for generation plant operation. This lighting should not be considered adequate to supply the requirements of 3.11.8.

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3.11,10 Electric motors. - Electric motors shall meet all the quality and performance requirements of CC-M-636 Or CC-M-641, as applicable.

3.11,11 Motor size. - The size of each electric motor shall be such that the driven load will not exceed continuous duty rating of the motor when the equipment is operated under the conditions specified herein. .

3.11.12 Motor service rating. - Electric motors shall be constructed to operate continuously at full rated load under the conditions specified herein without overheating or damage.

3.12 Instruments and controls, . All valves, indicators, gages, and controls required for operation of' the total system shall be supplied. A simple and logical system of control and indicator identification shall be employed which designates the location, function, fluid concerned and associated controls and indicators to each control or indicator, The nomenclature shall be consistently used in schematics, . diagrams, control identification, labels, and instruction manuals. Controls shall be so arranged to enable operation of the plant by one man. The control panel for the air source assembly and the liquefier assembly primary control panel shall be adjacent and arranged for consecutive operations during the normal operating modes. The system shall have all controls, both automatic and manual, required for safe, convenient, stable operation during start-up, steady state operating, and shut down. All instrumentation required to indicate operating conditions shall be provided. The controls and indicators shall be consecutively arranged in functional groups for normal operation to afford maximum efficiency. Instrumentation, including pyrometers and gages, shall have provision incorporated to protect them from damage due to exposure to temperatures and pressures beyond their normal range.

3.12.1 Interlocks. - Interlocks shall be provided throughout the system to minimize possibility of operator error and to prevent damage to the plant or injury to the operating or maintenance personnel.

3.12.2 Thermocouples. - Dual thermocouples shall be installed in any location where a single thermocouple failure would cause shutdown of the equipment. All thermocouples shall be properly shielded and protected from spurious electromotive force.

3.12.3 Valves. - All process valves used for throttling or modulating shall be equipped with position indicators which give the valve position of rotation and indicate total complete turns and parts thereof, All valves used in the system shall utilize packing compatible to liquid oxygen. Valves for low temperature fluids shall be of the extended stem and bonnet type. Both the valve operating handle and the valve stem packing shall operate at or near ambient temperature to insure prevention of injury to the operator or freezing of the valve. The valves shall operate through any cool-down-thaw sequence without malfunction. The valves located in the cold box shall have a removable seat and disc. Provisions shall be made to permit repair of these valves without removing the insulation material from the cold box other than a packing adjacent to the valve stem and bonnet.

3.12,3.1 Valve handles. - Each manually operated control or shutoff valve shall be provided with a tee, cross; or other easily turned type handle with a diameter of not less than 3 tithes per inch of valve nominal size, but not less than 1 inch.

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3.12.3.2 Valve cycling and leakage performance. - When closed with a torque of 60 ±5 pound-inches per inch of nominal size, and with the valve body at either ambient temperature or the atmospheric pressure boiling temperature of liquid nitrogen or liquid oxygen, the shut-off valve shall not leak more than 2 cubic inches of free air, oxygen, or nitrogen gas per hour per inch of nominal size from 50 psig to a downstream pressure of 0 psig. The valve shall be capable of not less than 2,000 cycles of operation (see 6.3.4) when subjected to a differential pressure Of 50 psig while closed, including at least 1,000 cycles at ambient temperature and 1,000 cycles with the valve body at the atmospheric pressure boiling temperature of liquid nitrogen or liquid oxygen without adjustment, repair, or maintenance, and without the above specified leakage rate being exceeded before, during; or after the cycling.

3.12.3,3 Valve overtorque performance. - The valve shall be designed and constructed to withstand an opening or closing torque of 300 ± 10 pound inches per inch of nominal size, with the valve body at either ambient temperature or the atmospheric pressure boiling temperature of liquid nitrogen or liquid oxygen. Such operation through 10 complete opening and closing cycles shall not result in a parts failure or a leakage rate in excess of that specified in 3.12,3,2 when the valve is subsequently opened and then closed with a torque of 60 ± 5 pound-inches per inch of valve nominal size.

3.12.3.4 Valve material. - The valve shall be fabricated from corrosion-resistant material not adversely affected by extended contact with oxygen, nitrogen, argon, hydrogen, helium, moist air, water, or cleaning solvents such as trichloroethylene.

3.12.3.5 Valve packing. - The valve packing shall be of a material safe for use in contact with liquid or gaseous oxygen or nitrogen, and that will effectively seal the valve stem without causing difficulty of operation.

3.12.3.6 Valve mounting. - All valves used during operation or thawing of the generating plant shall be so mounted and installed that they can be readily opened or closed by, and their control handles will be readily accessible to, operators standing at the control panels. All valves shall be so mounted to supporting members or large size piping that force applied to the control handles in opening or closing the valves will not damage the generating plant piping, and so mounted that the operator will be protected from liquid spray if a fitting or coupling breaks or leaks.

3.12.4 Electrical controls. - Heavy-duty, magnetically operated contactbrs, overload protection devices, and undervoltage protection devices of applicable power-handling capacity shall be provided for each electric motor. Heavy-duty switches or controls and overload protected devices shall be provided for all other electrically operated components.

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3.12.4.1 Electrical control panel. - Insofar as practicable, electrical controls, switches, and protective devices, including motor starters, transformers, and circuit breakers shall be grouped into an electrical control panel installed at a location convenient to the operators and maintenance personnel. This electrical control center shall include a master shut-off and disconnect stitch to control all generating-plant electrical power, and which will shut down all electrical equipment except the lighting specified in 3.11.8 when open.

3.12.5 Position indicators. - Manually operated controls that must be closely regulated for generating plant operation shall be equipped with indicators which shall have graduations to permit accurate adjustment of controls to predetermined effective settings. Both the controlling element and the indicator shall be designed with a minimum of hysteresis Or slack to insure the controlling element and the indicator are in true alignment regardless of the direction of controlling operation.

3.12.5.1 Indicator lights. - Where visual indicators are desired, neon lights which are adequately shielded from outside glare shall be used.

3.12.6 Lubricant level indicators. - Each liquid lubricant reservoir or sump shall be equipped with a lubricant level sight glass securely mounted in a position where it will be readily visible to operating and maintenance personnel. The lubricating level sight glass shall be long or large enough to permit personnel to determine the lubricating level in relation to its normal recommended position.

3.13 Accessories. - The generating plant shall include but not be limited to the following accessories.

3.13.1 Product purity analyzer. - The plant shall incorporate a continuously indidicating product purity analyzer. The indicator shall provide a continuous readout of. both liquid oxygen and liquid nitrogen purity by percentage. The mounting of the analyzer shall be suitable to protect it from shock and Vibration as specified herein.

3.13.2 liquid product discharge transfer hoses. - Two vacuum-jacketed metal hoses shall be supplied. Each hose shall be at least 10 feet in length and shall be equipped with a vacuum-jacketed bayonet coupling on the generating plant end which will provide leak-tight connection of the hose to the appropriate liquid product discharge connection. The servicing end of the liquid product hose and the liquid nitrogen hose shall be as specified in the contract to be compatible with the user's receiving tankage (see 6.2). The evacuation port closure of the hoses shall be of the pinch-off, weld-off type. Both hoses shall have safety features as specified in 3.5.7.2.

3.13.3 Fire extinguishers. - Two fire extinguishers and mounting brackets shall be provided as specified in the contract (see 6.2). Provisions shall be made for location of the extinguishers, one on either side of the plant, when set up for operation. Mounting brackets-shall be provided for securing the fire extinguishers within the equipment storage compartment when the plant is not in use.

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3.13.4 Operator weather protection. - A fabric enclosure shall be provided for personnel protection, It shall enclose and protect the operating area on all four sides, the fourth side being a skirt covering the open areas about the wheels and lowered panel under the operator's control console, The cover shall be of 5.5-oz. base nylon, neoprene coated, olive drab, having a finish weight of 17 to 18 ounces and conforming in style to MIL-C-20696, type II, class III; The enclosure shall be provided with a metal roof having a pitch of not less than 1/2 inch per foot down from the control panel. The enclosed clear floor area shall not be less than 4 feet by 10 feet and the average roof height shall be 7 feet. Attach points, cover supports, and cover tiedowns and anchors shall be provided, All three sides shall have windows with cover flaps with one of the sides also providing a means for entering and exiting the enclosure, All. three sides shall be capable of being rolled up to the roof individually and there shall be provisions for maintaining each side rolled up at the roof level, All sides shall have a tab along the bottom length to permit it to be buried into the ground and thereby increase the weather protection,

3.13.5 Jacks. - A minimum Of four mechanical jacks shall be provided to level and maintain in a level position the plant prior to operation, A minimum of two jack handles shall be provided. Each jack handle shall be a combination jack handle and lug wrench.

3.13.6 Special operation and maintenance tools. - When specified (see 6.2), a complete set of all special tools other than common handtools required for proper operation and maintenance of the generating plant and its accessories shall be provided,

3.13.7 Compartments. -

3.13.7.1 Tool compartment. - A compartment for storage of all tools required for normal operation and maintenance of the generating plant shall be furnished. The compartment shall be weatherproof if not contained in an enclosure. Provisions shall be incorporated for locking the tool compartment with a padlock.

3.13.7.2 Handbook and record compartment. - A compartment shall be provided integral to each end item for containing the equipment operation handbook and the equipment maintenance and inspection record. The compartment shall be weather-proof if not contained in an enclosure, and a minimum size of 9 by 11-1/2 by 2 inches.

3.14 Intercommunication provisions. - An intercommunication system to provide hearing protection and permit intelligible voice communications between attending personnel during generating plant operations shall be provided as specified in the contract. Where required, all the cord attachment receptacles, the electrical wiring inside the generating plant enclosure interconnecting the cord receptacles and inter-communication powering system, and necessary power filtration system shall be provided. The electrical cord attachment receptacles shall be installed where they will be pro-tected from rain or other precipitation during the operation or storage of the plant (see 6.2).

3.14.1 Part numbering of interchangeable parts. - All parts having the same manufacturers part number shall be functionally and dimensionally interchangeable, The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

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3.15 Dimensions. - When prepared for movement, the overall dimensions of the trailer shall not exceed the following and the additional dimension shown on figure 1 shall apply.

(a) Length: 15 feet (not including tow bar or tongue)
(b) Width: 8 feet (excluding running lights)
(c) Height: 8 feet and 6 inches

3.16 Weight. - The total gross weight of the generating plant shall not exceed 15,000 pounds when prepared for storage or transportation.

3.17 Finishes and protective coatings. -

3.17.1 Finishes. - The use of any protective coating that will crack, chip, or scale with age or exposure to environmental conditions specified in 3.6.2 shall be avoided. When materials used in the construction are subject to corrosion in moist salt air or other environmental condition likely to occur during normal service usage, they shall be protected against corrosion in a manner that in no way prevents compliance with the requirements of this specification,

3.17.2 Exposed parts and surfaces, - All exposed metal parts and surfaces with the exception of the parts and surfaces that. contact the high purity product and certain corrosion-resistant materials, shall be treated as specified in MIL-STD-171.

3.17.2.1 Corrosion-resistant metal surfaces. - Corrosion-resistant metal heat exchange surfaces and corrosion-resistant metal pipes and tubes carrying cryogenic liquid-or cold gases shall not be painted.

3.17.3 Surfaces contacting high purity product. - All surfaces, parts, and fittings of the generating plant that contact the high purity product shall be cleaned with a nonflammable solvent, such as trichlorethylene, to remove all metal chips, oil, grease, and other foreign material. No other cleaning, priming, or painting with organic materials shall be performed on these surfaces.

3.17.4 General cleaning instructions. - Following completion of Fabrication and assembly operations, the generating plant shall be thoroughly cleaned and degreased to remove dirt, excess soldering; brazing and welding flux, welding slag, scale, loose, spattered, or excess solder, metal chips, loose or chipped paint, spilled chemicals, and other foreign material. Cleaning shall be continued until visual inspection shows no evidence of contamination or foreign matter, ultraviolet inspection of surfaces contacting high purity product shows no evidence of fluorescence and particulate contamination of the product withdrawn does not exceed the following:

(a) When the liquefaction plant is filled to capacity with filtered liquid nitrogen and allowed to set for not less than 2 hours, the residue remaining after evaporation of the first liter withdrawn through the product discharge line shall contain:

(1) No solid particle with any dimensions greater than 1,000 microns
(2) No fibrous particle with a length greater than 6,000 microns
(3) No more than 25 milligrams total of both solid and fibrous particles

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3.17.4.1 Decreasing. - All surfaces, parts, and fittings of the generating plant that will be in contact with the high purity product shall de degreased by one of the following methods. Precautions shall be taken to insure that solvents do not contact parts fabricated from incompatible materials.

(a) Method A - Vapor decreasing process, using stabilized trichloroethylene conforming to O-T-634 in a standard commercial vapor degreaser or by blowing decreasing vapors into the component in such a manner that the vapor will condense on and properly clean all surfaces requiring decreasing. Operation of a commercial - vapor degreaser shall be in accordance with the manufacturers recommendations. Following the vapor decreasing treatment, all solvent shall be removed by baking in an oven; by purging with hot, dry oil-free air or nitrogen; or by vacuum evacuation,

(b) Method B - Solvent decreasing process, using stabilized trichloroethylene or commercial oxygen- and nitrogen-safe cleaning solvent at ambient temperature to thoroughly wash all surfaces requiring degreasing. If the solvent contains more than 1 percent oil or hydrocarbons after the cleaning and being drained from the component, the decreasing operation shall be repeated with clean solvent. Following the decreasing process, all solvent shall be removed by baking in an oven; by purging with hot, dry, oil-free air or nitrogen; or by vacuum evacuation,

(c) Method C - Detergent decreasing process, in which the components to be cleaned are washed with hot inhibited alkaline cleaner until free from oil, grease, and other contaminant materials. Following this treatment, all surfaces (internal and external) shall be rinsed thoroughly with fresh, clean, hot water and dried by blowing with filtered and dried oil-free air or nitrogen, or by baking at a temperature of 250° to 300° F until all water is removed.

(d) Method D - Steam cleaning by subjecting the component to a stream of steam only or to a stream of steam with an added cleaning compound followed by steam alone. Cleaning compound shall conform to P-C-437. Following steam cleaning, all surfaces, internal and external, as applicable, shall be purged with hot, oil-free air or nitrogen or by baking at a temperature of 250° to 300° F until all water is removed.

(e) Method E - Detergent cleaning with four percent detergent solution (6 ounces of detergent with one gallon of water) for 30 minutes at 120° F. Detergent shall conform to MIL-D-16791. This procedure is applicable to plastics, O-rings, packing, and similar materials, After cleaning, rinse with distilled water several times, Dry by purging with hot, oil-free air or nitrogen.

(f) Method F - Flushing or rinsing parts and components with oxygen systems cleaning compound conforming to MIL-C-8638. After cleaning, parts and components shall be dried with hot, oil-free air or nitrogen.

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3.17.5 Petroleum and other flammable solvents shall not be used on such surfaces,

3.17.6 All cleaning materials and solvents shall be thoroughly removed from the plant components following the cleaning and decreasing processes.

3.17.7 Lubrication. - All generating plant machined bearing surfaces that are normally lubricated in operation and that do not come into contact with the contained product shall be thoroughly lubricated with the recommended military specification or federal specification lubricants to be ready for immediate operation,

3.18 Colors. - Generating plant exterior parts and surfaces shall be painted in fied by contract (see 6.2).

3.19 Operation markings. -

3.19.1 Marking. - Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.19.1.1 Component maintenance identification. - All components subject to preventive maintenance, replacement or repair shall be permanently marked in a manner to be in full view to the technician in the course of his maintenance procedures.

3.19.1.2 Danger zone identification. - Adequate signs or markings shall indicate danger zones such as the turbine exhaust or other areas where inadvertent action might cause damage, personal injury, or a shut down of the plant operation.

3.19.1.3 Lubrication point marking. - Each lubrication point shall be identified by a filled-in circle of approximately 2-inch diameter around the lubrication point. The circle shall be painted on the equipment with gloss enamel conforming to TT-E-489. The color shall be insignia red, color No. 11136 of FED-STD-S95: Lubricant filler caps shall be painted with the same enamel.

3.19.1.4 Lubricant level sight glass marking. - Lubricant level sight glasses shall be plainly marked to show the lubricant level recommended for operation of the generating plant.

3.19.1.5 Lubrication warning plates. - A weatherproofed red warning plate with white lettering shall be affixed to the generating plant in a conspicuous location at each point where the use of high-pressure lubricating equipment will cause damage to grease seals or other parts.

3.19.2 Placards. - A flow diagram, an electrical system diagram, a transportation data plate weight, dimensions and center of gravity plates, instruction plates, identification plates, and warning plates shall be supplied.

3.19.2.1 Stenciled instructional markings. - The following instructional markings shall be stencil-led on the generating plant as specifiedo The stencil material color shall be as specified by the procuring activity.

3.19.2.1.1 Tire pressure. - The pressure to which the generating plant running gear tires should be inflated shall be marked with 1/4-inch high block letters in a protected and readily visible location near each wheel.

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3.19.2.1.2 Fuel type end tank capacity. - The type of fuel to be used and the fuel tank capacity in U. S. gallons shall be marked with block letters not less than 3/16 inch high in a readily visible location adjacent to the fuel filler cap or connection,

3.19.2.1.3 Lubricant type and capacity. - The type of lubricant to be used and the refill capacity of the lubricant sump shall be marked with block letters not less than 3/16 inch high in a readily visible location adjacent to each lubricant filler cap."

3.19.3 Name plate location. - The generating-plant nameplate shall be securely attached to the outside of the trailer body in a location near the control panel and where it will be readily visible during operation, storage, and transport of the equipment.

3.19.4 Component marking. - All operator control equipment, such as pushbuttons, switches, valves and gages, shall be identified as to function by an identification plate with permanently imprinted lettering visible without removal or disassembly, where possible. In addition, all process control equipment shall be further identified by its schematic designation,

3.19.5 Transportation data plate. - The transportation data plate in accordance with MIL-P-514 shall be securely attached to the outside of the trailer body in a location near the control panel where it will be readily visible and accessible to transportation personnel during generating storage and shipment. The transportation data plate shall include the following information:

TRANSPORTATION DATA PLATE

Shipping weight	Overall width
Shipping cubage	Overall height
Overall length	C.G. Location

T.O. or T.M. Number

3.19.6 Lifting instruction date. - An instruction plate containing all information necessary for transportation personnel to:

(a) Safely lift and move the generating plant with a single, high-capacity, forklift truck
(b) Safely lift and move the generating plant with multiple, smaller (down to 4,000 pounds) capacity, forklift trucks, and
(c) Rig a safe lifting sling from common cable and spreader bar components and safely lift the generating plant with a 2g acceleration which shall be securely attached to the outside of the control housing assembly near the transportation data plate. The lifting instruction plate shall be of the same material and prepared in the same manner as the transportation data plate.

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3.19.7 Color marking. - The indication scale of each instrument (pressure gage, differential pressure indicator, temperature indicator, and liquid level indicator) used for normal operation and control of the tank shall be permanently and plainly marked; with green and red to show the proper operating and the other-than-proper-operating or danger zones, respectively.

3.19.8 Insulation identification daze. - An insulation identification plate containing all information necessary to completely identify the insulation material and showing the quantity in pounds of the insulation material installed in the cold box insulation space shall be permanently affixed in a readily accessible and protected location that will be within the control housing assembly when it is closed for transport or storage. The insulation identification plate shall be prepared in the same manner as the operation and precautionary instruction plate.

3.19.9 Identification of product. - Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.19.10 Part numbering of interchangeable parts. - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract. or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the rignt to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Examination and testing of the generating plant shall be classified as follows:

(a) Reproduction inspection (4.4)
(b) Quality conformance inspection (4.5)

4.3 Test conditions. - Unless otherwise specified, tests shall be performed under ambient conditions (see 6.2 and 6.3.1).

4.3.1 Apparatus. - Whenever practical, apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at intervals properly spaced to insure continued laboratory accuracy.

4.3.2 Tools. - Special tools and field equipment shall be used to the maximum practicable extent during testing. Utilization of these tools and of the field equipment shall. be sufficient to determine their usefulness. Instances where the special tools and field equipment furnished if required are inadequate shall be recorded in detail.

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4.3.3 Instrumentation. -

4.3.3.1 Pressures. - Pressures and pressure differentials shall be measured by means of the pressure gages and differential pressure meters furnished as part of the generating plant.

4.3.3.1.1 Pressures in excess of atmospheric pressure shall be recorded in pounds per square inch gage. Pressures up to and including atmospheric pressure, shall be recorded in either millimeters of mercury or inches of mercury absolute. Differential pressures shall be recorded, in inches of water or inches of mercury, as applicable.

4.3.3.1.2 Barometric pressures shall be measured by a properly calibrated mercurial barometer and reported in millimeters of mercury absolute.

4.3.3..2 Temperatures. - Temperatures shall be measured by means of appropriately located thermometers (including vapor-pressure thermometers), by thermocouples used with calibrated potentiometers, or by resistance bulbs, as applicable. The temperature indicating instruments furnished as part of the generating plant shall be used whenever practicable.

4.3.3.2.1 Temperatures shall be recorded in degrees Fahrenheit.

4.3.3.3 Rotational speeds. - Rotational speeds shall be determined by a revolution counter, by a tachometer, or by a strobotachometer if calibration data can be furnished to attest to its accuracy. Whenever practicable, the rotational speed indicating instruments furnished as part of the generating plant shall be used.

4.3.3.3.1 Rotational speeds shall be recorded in revolutions per minute.

4.3.4 Production rate determination. - Liquid oxygen and liquid nitrogen production rates shall be determined by weighing the liquid delivered into a container maintained at atmospheric pressure and through the generating plant liquid product hoses. Throughout a production capacity measurement period, the column liquid levels and generating plant pressure shall be maintained within the limits established for stable operation,

4.3.4.1 Collection container calibration. - Containers for collection and weighing of liquid product should be calibrated for heat leak evaporation losses. The weight of liquid found to evaporate from the container in a length of time equal to the production collection period may be added to the weight collected to determine the true product delivery capacity, Heat leak evaporation loss rate tests shall be made only when the container is not less than half full, and when the filled container has set for not less than 12 hours after filling to permit stabilization of temperatures in the container and its insulation. Allowance shall not be permitted for liquid evaporated in the liquid product servicing system, including the delivery hoses, since this system is part of the generating plant.

4.3.4.2 Production rate reporting. - Liquid production rates shall be reported in pounds per hour of the liquid actually delivered into the weighing container.

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4.3.5 Fuel. - The power source fuel used for the testing specified herein shall be as follows.

4.3.5.1 Fuel procurement. - Unless otherwise specified by the contract or purchase order, grade JF-4 fuel conforming to MIL-T-5624 shall be used for testing generating plants procured by the Air Force.

4.3.5.2 Fuel consumption. - Fuel consumption shall be determined by measuring the quantity necessary to refill a reservoir after a suitable elapsed time, or by measuring the drop of fuel level over a suitable elapsed time in a container of accurately known dimensions. Fuel consumption shall be reported as the gallons at 600 F of fuel consumed per operating hour.

4.3.6 Lubricants. - Lubricants required for the testing specified herein shall be in accordance with applicable military specifications or federal specifications and of the types and grades recommended by the equipment manufacturer,

4.3.6.1 Lubricating oil consumption. - Lubricating oil consumption rates shall be determined by measuring the quantity necessary to maintain a given level in the oil reservoir. Lubricating oil consumption shall be recorded as the number of quarts consumed per hour or the number of quarts consumed per operating hour, whichever provides more convenient units.

4.3.7 Observations. - During the progress of all testing, such characteristics as smoothness of operation, vibrations, ease of control, speed regulation, pressure regulation, freedom from leaks, and the operation of auxillary equipment shall be carefully observed and recorded to determine whether the generating plant conplies with all requirements specified herein.

4.3.8 Preparation for test. - The generating plant shall be prepared for test by assembling, setting up, adjusting, and servicing as specified in the operation instruction handbook.

4.3.9 Test data. -

4.3.9.1 Correction of data. - If the tests specified herein cannot be performed under the specified conditions the test may, upon specific approval by the procuring activity, be performed under other conditions satisfactory to the procuring activity and the performance under the specified conditions calculated from the test results obtained. The manner of calculation shall be demonstrated, and actual test data proving the correctness of the calculation methods shall be presented for review by the procuring activity.

4.3.9.2 Operational data. - Operational data shall be recorded at intervals not to exceed 1/2 hour in performance demonstrations and other short tests and at intervals not to exceed 1 hour in endurance runs. The data recorded shall include the following, as applicable to the particular test being performed:

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(a) Air supply equipment:

(1) Discharge air pressure (PSI)
(2) Discharge air temperature (°F)
(3) Lubricating oil pressure (PSI)
(4) Lubricating oil temperate (°F)
(5) Fuel pressure (PSI)
(6) Power unit speed (RPM)
(7) Power unit exhaust temperature (°F)
(8) Compressor speed (RPM)
(9) Hourmeter reading
(10) Ambient air temperature (°F)
(11) The other operating speeds, flow rates, liquid lubricant levels, pressures, temperatures, instrument readings, and control settings used in operating the air supply equipment and determining whether it is operating properly.

(b) Air liquefaction and separation equipment:

(1) Process air supply pressure (PSI)
(2) Process air supply temperature (°F)
(3) Column liquid levels (inches of liquid)
(4) Column pressures (PSI) (above liquid product}
(5) Air pressure to exchangers or regenerators (PSI)
(6) Air temperature to exchangers or regenerators (°F)
(7) Air pressure from exchangers or regenerations (PSI)
(8) Air temperature from exchangers or regenerators (°F)
(9) Air temperature to refrigeration equipment (°F)
(10) Air temperature from refrigeration equipment (°F)
(11) Expander speed (RPM)
(12) The other operating speeds, flow rates, liquid levels, pressures, temperatures, instrument readings, and control settings used in operating the air liquefaction equipment and determining whether it is operating properly.

4.3.9.3 Production data. - Production data (average rate at which liquid oxygen and liquid nitrogen were produced during the immediately preceding period since the prior rate determination) shall be reported not to exceed 2-hour intervals during generating plant liquid production,

4.3.9.4 Psychometric data. - Psychometric readings and the relative humidity shall be reported at intervals not to exceed 4 hours during generating plaint operation,

4.3.9.5 Barometric pressure. - The barometric pressure shall be recorded at intervals not to exceed 4 hours during generating plant operation.

4.3.9.6 Fuel and lubricant consumption. -

4.3.9.6.1 The average fuel consumption rate for each hour of operation shall be recorded.

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4.3.9.6.2 The average lubricating oil consumption rates for each 24-hour or shorter period of operation shall be recorded at the end of the period.

4.3.9.7 Tolerances. -

4.3.9.7.1 Data on gage pressures measured in the range from atmospheric to 10 psi shall be accurate to within 1/2 psi.

4.3.9.7.2 Data on gage pressures measured in the range from 10 to 50 psi shall be accurate to within 1 psi.

4.3.9.7.3 Data on gage pressures in excess of 50 psi shall be accurate to within 2 percent of the actual numerical reading.

4.3.9.7.4 Data on pressure differentials shall be accurate to within 2 percent of the actual numerical reading.

4.3.9.7.5 Data on absolute pressures (including barometric pressures) measured in the range from 1 mm. Hg to and including atmospheric shall be accurate to within 0.1 mm. Hg.

4.3.9.7.6 Data on weights and liquid flow rates shall be accurate to within 1 percent of the actual numerical reading.

4.3.9.7.7 Data on gas and vapor flow rates shall be accurate to within 2 percent of the actual numerical reading.

4.3.9.7.8 Data on rotational speeds shall be accurate to within 2 percent of the actual numerical reading.

4.3.9.7.9 Data on both dry-bulb and wet-bulb temperatures shall be accurate to within 2° F.

4.3.9.7.10 Data on product purities shall be accurate to within 0.1 percent.

4.3.9.8 Pressure test record. - A record of the pressures at which each generating plant component is tested and the length of time during which it is subjected to the pressure shall be recorded.

4.3.10 Preliminary run-in. - The nature and extent of running-in shall be determined by the manufacturer, and shall be performed prior to the testing as specified herein. All necessary adjustments other than normal control adjustments shall be made during this run-in and shall remain undisturbed thereafter._

4.3.11 Inspections, servicings, and adjustments. - Inspections, servicings, and control or other adjustments shall be performed as specified by the operating, servicing, and maintenance instruction manuals unless such instructions are contrary to those necessary for compliance with the requirements specified herein.

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4.4 Preproduction inspection. - Preproduction inspection shall consist of all the tests specified in 4.6.

4.4.1 Preproduction sample. - Unless otherwise specified by the contract or purchase order, one complete generating plant shall be subjected to preproduction inspection specified in 4.4.3.

4.4.2 Inspection report. - Following completion of the preproduction inspection, a test report which includes all tests recorded as specified herein shall be prepared in accordance with MIL-STD-831 and shall be furnished to the procuring activity as specified in the contract or purchase order.

4.4.2.1 Reliability and maintainability information. - The following information shall be included as an appendix to the preproduction inspection reports:

(a) A description of and the appropriate information below concerning any tests that were performed prior to those described in the test report and that were intended to apply to the contract.

(b) All failures, maintenance actions, servicing, irregular functioning, and other similar events of the testing. Such actions and events shall be identities by a brief description of the action or event and by the accumulated operating time, cycles, miles, or position in the test procedure, as appropriate. Test conditions during failures or irregular functioning shall be reported.

(c) Test operator and maintenance technician actions, test equipment and test facility failures, and other events that might act as grounds for a request that an equipment failure not be counted as a reliability failure. Detailed description of the events and the analyses to substantiate any such requests made shall be included, and shall be clearly cross-referenced to each applicable failure.

(d) A summaryof the engineering analysis and of any tests performed to determine assignable causes for any failure or irregular functioning.

(e) A summary of the engineering analysis leading to any correction made to design, construction, quality control, or other procedures, or leading to any corrections recommended to be made to production items, or proposed to be made. The summary shall also include an analysis of the predicted effectiveness of such corrections. . Failures that have been corrected by design changes or by other means shall be counted as reliability failures until the corrections have been both analyzed and verified by test sufficiently to substantiate the effectiveness of the correction to the satisfaction of the procuring activity.

(f) The clock the and man-hours required for each maintenance and servicing action performed during the testing. Only the time needed for actually preparing for and performing the tasks shall be measured and reported. A brief description of the experience and qualifications of the personnel taking such actions shall be included. This information shall include a summary of the data resulting from the servicing and maintenance tests required by 4.6.13. In addition, the clock "time and the man-hours required for actual access and disassembly shall be reported for any teardown inspections done for mechanical checks in accordance with 4.6.16. The number and description of parts and assemblies removed and causing interference shall be noted for each teardown and inspection made. Administrative time, such as filling out records and logistics time, such as obtaining parts from stock, shall not be included.

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(g) Estimates of the man-hours anticipated for other servicing and maintenance actions anticipated in field use. A brief description of the qualifications and skills required to accomplish these servicing and maintenance actions shall also be included,

(h) Test activity or contractor comments on item features or requirements that, if modified, should improve the usefulness of the item.

(i) Test activity or contractor comments on use or maintenance conditions to be cultivated or avoided to increase the reliability and useful life Of the item.

(j) Supporting engineering data and calculations to substantiate compliance with the center-of-gravity requirements of 3.5.4.4.

(k) Supporting engineering data and calculations to substantiate compliance with the air transport flight and taxiing acceleration requirements of 3.5.9.3.

4.4.2.1.1 Any of the above information that is already included in the test report body or in other documents submitted to the procuring activity need not be repeated in the information required by this paragraph, but clear reference to the location and to the date of submission of the data shall be included.

4.4.2.1.2 All table and column headings and all abbreviations and symbols used shall be clearly defined where used or in a table of abbreviations and symbol definitions placed in the front part of the appendix.

4.5 Quality conformance inspection. - Quality conformance inspection shall consist of the following:

(a) Individual tests	(4.5.1)
(b) Sampling plans and tests	(4.5.2)

4.5.1 Individual tests. -

4.5.1.1 Spring-loaded relief valve test. - Each spring-loaded relief valve shall be subjected to the test specified in 4.6.1.

4.5.1.2 Generating plant tests. - Each generating plant shall be subjected to the following tests as specified in 4.6:

(a) Examination of product	(4.6.3)
(b) Pressure test	(4.6.4)
(c) Cleaning effectiveness tests	(4.6.5 through 4.6.5.3.1)
(d) Performance demonstration	(4.6.7 through 4.6.7.3)
(e) Road operation	(4.6.6.1.1)

4.5.2 Sampling plans and tests. -

4.5.2.1 Bursting disc sampling tests. -

4.5.2.1.1 Bursting disc lot. - A bursting disc lot shall consist of discs manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

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4.5.2.1.2 Bursting disc sampling and test. - Three of each size, design, and construction bursting disc from each lot of 50 or less shall be selected at random and each disc subjected to a gradually increasing clean, dry, oil-free air or nitrogen pressure until it ruptures.

4.5.2.1.3 Bursting disc rejection and retest. - In case a disc fails at a pressure outside the limits established for compliance with this specification, five additions sample bursting discs selected at random from the seine lot shall be tested in the same manner. If more than one of the original three sample bursting discs or any of the five additional sample bursting discs fails at a pressure outside the specified limits, the entire lot shall be rejected,

4.6. Inspection methods. -

4.6.1 Spring-loaded relief valve test. - Prior to installation on the generating plant, each spring-loaded relief valve shall be subjected to a gradually increasing pressure of clean, dry, oil-free air or nitrogen applied to its inlet until it starts to discharge. The pressure shall then be reduced until the valve completely reseals, This procedure shall be repeated until the valve has started to discharge, and resealed, not less than three times. Following this testing, the valve shall be checked for leakage by application of a soap film (film across the outlet and over all outside surfaces) with not less than the maximum working pressure of the component which the valve will be used to protect, being applied to the valve inlet. Failurs to open or reseal within the specified limits or an indication of leakage during the soap film test shall be cause for rejection.

4.6.2 Shutoff and control valve testing. - Each valve shall be subjected to the following tests in the order specified.

4.6.2.1 Cycling test. - The valve shall be closed with a torque of 60 ± 5 pound inches per inch of nominal size, and subjected to a compressed air-or nitrogen gas inlet pressure of not less than 1-1/4 times its maximum working" pressure with the outlet open to atmospheric pressure. The valve shall then be opened to not less than its 3/4 open position, after which it shall be again closed with a torque of 60 ± 5 pound inches per inch of nominal size and the specified inlet pressure reestablished. This shall constitute one cycle. One thousand such cycles shall be accomplished without intervening valve lubrication, adjustment, or repair while the valve body is subjected to ambient temperature. The 100-cycle test shall then be repeated with the valve body submerged in liquid nitrogen. Damage to or improper operation of the valve as a result of this test shall be cause for rejection. A valve cycle test shall also require the valve to be closed with a torque of 60 ± 5 pound inches per inch and the valve is then to be heated to 300°F for a period of 30 minutes, Then cool the valve to liquid nitrogen temperature and-proceed to open the valve by hand.

4.6.2.2 Overtorque test. - At the conclusion of the cycling test specified in 4.6.2.1, the valve shall be removed from the liquid bath and permitted to stabilize at ambient temperature. The valve shall then, without intervening, lubrication, adjustment, or repair, be fully opened, and then closed with a torque of 300 ± 10 pound inches per inch of nominal size. This opening and closing shall constitute one cycle. Ten such cycles shall be accomplished without intervening valve lubrication, adjustment, or repair. Damage to or improper operation of the valve as a result of this test shall be cause for rejection.

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4.6.2.3 Leakage test. - Following the overtorque test specified in 4.6.2.2 and without intervening valve lubrication, adjustment, or repair, the valve snail be fully opened, and then closed with a torque of 60 ± 5 pound inches per inch of valve nominal size. The valve shall then be completely immersed in clear water with the inlet subjected to a compressed air or nitrogen gas pressure of not less than 1-1/4 times its maximum working pressure for not less than 10 minutes with the outlet open. Any leakage past the seat, through the packing, or through the valve body shall be collected and measured, Total collected gas leakage exceeding 2 cubic inches of free air or nitrogen gas per hour per inch of valve nominal size shall be considered cause for rejection.

4.6.3 Examination of product. - The generating plant shall be carefully examined, including mechanical inspection of all components and parts, to determine whether it complies with the requirements specified herein. Particular attention during the examination shall be paid to the following factors:

(a) General design
(b) Component arrangement
(c) Alignment of parts and components
(d) Accessibility to parts and components
(e) Ease of performing adjustments-ad control actions
(f) Free and proper functioning of all parts
(g) Quality of materials and wokmanship
(h) Finish and appearance
(i) Marking

All pertinent data concerning condition, defects of manufacture, damage in transit, and damage through use prior to test shall be reported. This inspection shall include servicing of the generating plant for use.

4.6.3.1 Dimensions and weight. - The overall dimensions and total weight, including all equipment and accessories, of the completely assembled generating plant, ready for shipment or setting up for uses shall be determined and reported. The dimensions and weight shall not exceed the limits specified in 3.15 and 3.16.

4.6.4 Pressure test. - Each component and circuit of the generating plant that operates under positive pressure in normal service shall be pressure tested in accordance with the procedures and requirements established in Section VIII of the ASME Boiler and Pressure Vessel Code. Automatic pressure relief devices need not b-e installed for this test. Any distortion, damages or leaks resulting from the pressure testing shall be cause for rejection of the components affected,

4.6.5 Cleaning effectiveness tests. -

4.6.5.1 Visual inspection. - All equipment, pipes, and components that will contain or carry high purity oxygen or nitrogen shall be visually examined where possible for evidence of corrosion products, metal chips, mill scale, weld scale, oil, grease, paints, preservatives, decals, or other contamination or foreign matter. Any evidence of contamination or foreign matter shall require recleaning and retest.

4.6.5.2 Ultraviolet light inspection. - All accessible surfaces and parts, including insofar as practicable the internal surfaces of the components and lines, that will contain, contact, or carry high-purity oxygen or nitrogen shall be visually inspected for hydrocarbons with ultraviolet light. Any evidence of fluorescence shall require recleaning and retest.

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4.6.5.3 Particulate matter testing. - Throughout the performance demonstration and endurance testing performed on the generating plant, both the liquid oxygen product and the liquid nitrogen product shall be passed through a filter conforming to 4.6.5.3.1 before being filled into the receiving containers. Following a test period, the filter shall be removed from the product delivery line, thoroughly cleaned, and the contaminant material removed by the filter from the product stream collected, weighed, and examined for particle size. The generating plant components and piping shall be cleaned and the test repeated until the particulate matter collected contains:

(a) No solid particle with any dimension greater than 1,000 microns
(b) No fibrous particle with a length greater than 6,000 microns
(c) Not more than one milligram total of both solid and fibrous particles per each 25 gallons of liquid product discharged through the product delivery system.

4.6.5.3.1 Particulate matter collection filter. - The particulate matter collection filter element shall be constructed from stainless-steel or monel. Sintered or powdered metal materials shall not be utilized unless backed up by a wire-wound or wire-mesh sloth element, in which the wires have been fused under controlled conditions, so as to positively prevent migration of parzi.ties that may be shed by the sintered or powdered metal portion. The filter element shall be fused or otherwise fastened to the housing in a manner that will insure against bypass of the element, and shall:

(a) Remove not less than 98 percent by weight of all particles whose smallest dimension is 10 microns or greater (10 micron nominal rating).
(b) Remove all particles whose smallest dimension is 40 microns or greater (40 micron absolute rating).

4,6,6 Transportability and handling tests. -

4.6.6.1 Mobility. - The generating plant shall be tested for mobility characteristics in accordance with MIL-M-8090 applicable to equipment items having type III, group C, mobility. After each test, the generating plant shall be examined for misalignment, distortion, and other defects that may be caused by the mobility testing to determine whether the equipment was adversely affected by the testing. The generating plant shall perform in accordance with MITLA!-8090 or the generating plant shall be considered to have failed the testing.

4.6.6.1.1 Individual article road operation. - The generating plant shall be towed riot less than 25 miles over paved highways or similar improved road (see 6.3.6) and not less than 10 tiles over graded gravel or similar unimproved roads (see 6.3.7). The vehicle speed shall be maintained at not less than 50 mph on the improved roads and at not less than 10 mph on the unimproved roads wherever practicable. Not less than five sudden stops shall be made in which the generating plant and towing vehicle are brought to a complete stop from a speed of not less than 40 mph within a maximum distance of 100 feet. Throughout the road operation, the generating plant shall be closely observed as to its trackability, ease of handling, tendency to yaw, tendency to sway, et cetera. Upon completion of the road handling and operation testing the generating plant shall be carefully inspected for damage, misalignment, distortion) or any other deterioration sustained as a result of the test operation, Undesirable readability features or any damage resulting from the test which would affect its use and operation shall be cause for rejection. A complete description of any adverse handling characteristics or damage sustained shall be reported.

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4.6.6.2 Air transport flight and taxiing acceleration forces. - The completely assembled generating plant ready for shipment shall be subjected to the flight and taxiing acceleration force loads specified in MIL-A-8421. Failure to satisfactorily pass the specified testing for demonstrating compliance with the requirements of 3.5.9.3 shall be cause for rejection.

4.6.6.3 Vibration. - The completely assembled generating plant ready for shipment shall be subjected to the vibration testing specified in Method 514, procedure X, of MIL-STD-810 for category g equipment, except that the generating plant need not be operated during any part of the Vibration testing.

4.6.6.3.1 During each phase of the testing, the generating plant and the test instrumentation shall be observed for resonance of piping, instruments, accessories, braces, brackets, and other installed parts, devices, and equipment; loosening or loss of threaded, riveted or other fasteners; and either damage to or indications of impending failure of the overall generatingplasm or any of its components, Full details or resonances, loosening of parts or fasteners, or other damage or indications of impending failure shall be reported or. the test data sheet. Programmed vibration testing not completed because of a failure or impending failure, including any remaining resonance dwell, shall be repeated after corrective action.

4.6.6.3.2 The generating plant shall be carefully examined after each test phase and following completion of the overall test. Fractures or bending of piping, braces, or brackets; fracture or failure of welded, brazed, or soldered joints, connections, or fittings; loosening or loss of parts or of threaded, riveted or other type fasteners; damage to or loss of calibration in gages and other instruments; or physical damage to equipments shall be considered a failure of the generating plant to pass the testing.

4.6.6.4 Hoisting and tiedown provisions. - Weights totaling not less than twice the generating plant shipping weight shall be so attached to the generating plant that the combined weight of the generating plant and the attached weights will be supported by the hoisting and tiedown fittings when the equipment is raised by an overhead crane. The generating plant and weights combination shall then be raised to a height of not less than 3 feet above the floor or ground by a single hook overhead crane, using the hoisting and tiedown provisions, held in that position for not less than 3 minutes, and then lowered to the floor or ground. This procedure shall be repeated three times. Any damage to or distortion of any generating plant component shall be cause for rejection.

4.6.7 Performance demonstration. - The generating plant shall be operated in accordance with the following procedures to determine and demonstrate its performance capabilities prior to undergoing endurance testing.

4.6.7.1 Liquid oxygen production test. - The generating plant shall be set up, as specified in the operation instruction handbook, thawed, started, and operated under the conditions specified in 3.7.1 until the rate at which it will continuously produce and deliver liquid oxygen conforming to 3.7.4.1 has been established. Liquid oxygen production shall be continued at maximum rate for not less than an additional 16 hours after rated production capacity has been achieved. The length of operating the from starting the power source to reaching the production rate specified in 3.7.1 and the average production rate maintained after initially reaching this rate shall be reported. The length of operation time required to reach the production rate specified in 3.7.1 shall not exceed the time specified in 3.7.2, and the average production rate achieved shall be not less than the rate required by 3.7.1, or the generating plant shall be considered to have failed the test.

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4.6.7.2 Liquid nitrogen production test. - Upon completion of the testing specified in 4.6.7.1, the generating plant shall be adjusted for liquid nitrogen production and operated under the conditions specified in 3.7.1 until the rate at which it will continuously produce and deliver liquid nitrogen conforming to 3.7.4.2 has been established. Liquid nitrogen production shall be continued at maximum rate for not less than an additional 16 hours after rated production capacity has been achieved. The clock time required to make the changeover from liquid oxygen to liquid nitrogen production, the length of operation time after changeover to reaching the production rate specified in 3.7.1, and the average production rate achieved shall be reported, The average production rate achieved shall be not less than the rate required by 3.7.1, or the generating plant shall be considered to have failed the test.

4.6.7.3 Performance demonstration test conditions, - Throughout the performance demonstration, operation of the generating plant shall be as continuous as possible. Production rate averages shall be computed on elapsed clock time with no allowances permitted for shut-downs or production breaks necessary for or caused by servicing, preventive maintenance, adjustments, repairs, or ether reasons. Throughout the performance demonstration, liquid product shall be delivered through the transfer hoses specified in 3.13.2 into tanks maintained at approximately atmospheric pressure and in which the liquid levels are not less than eight feet above the ground, floor, or surface on which the generating plant is operated.

4.6.8 Endurance testing. - Following completion of the performance demonstration testing specified in 4.6.7, the generating plant shall be subjected to an endurance test consisting of not less than four 240-hour production periods. This testing shall be performed as follows.

4.6.8.1 Preparation for test. - The generating plant shall be thawed, serviced, and prepared for normal operation as specified in the operation, servicing, and maintenance instruction handbook and in compliance with 3.6.4.

4.6.8.2 Operating conditions. - The four 240-hour production periods of the endurance testing shall be performed as specified and in the order listed below. Operation throughout each production period shall be as continuous as possible except that the generating plant may be shut down for no longer than 30 minutes during each 8-hour (clock-the) operational period, total shutdown time not to exceed 90 minutes in any one 24-hour operational period, for servicing, minor adjustments, inspections, preventive maintenance, and minor repairs. The length of and reason for each shutdown or interruption of production, the time to achieve rated production at the beginning of the test period and after each shutdown or interruption, corrective actions taken, and all difficulties and malfunctions encountered shall be reported,

4.6.8.2.1 If for any reason an interruption of greater length than that permitted by 4.6.8.2 is necessary and the shutdown time results in a production-break, (time during which product is not actually being delivered at equal to or better than rated capacity of 8 hours or less) the required 240-hours of production time for the period shall be extended by 2 hours for each hour of shutdown the plus the time required to resure production. (Example: A 5-hour shutdown requiring additional 3 hours to resume production would extend the required production the for the period to 253 hours of actually delivering product at not less than rated capacity,) If a production break of more than 8 hours is necessary, the purpose of the specified production period will be considered defeated, and the production period shall be repeated. Restart penalty may be waived by the procuring activity after 80 percent of the inspection has been completed.

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4.6.8.2.2 If during any of the eight-hour operational periods referenced in 4.6.8.2 it becomes necessary to interrupt operation other than during the scheduled 30-minute shutdown time permitted by 4.6.8.2 for servicing, adjustment, maintenance, or repair, no part of the production time accumulated during that 8-hour period shall be credited toward completion of the production period.

4.5.3.3 Liquid oxygen. production phase. - The generating plant shall be adjusted for production of liquid oxygen, started, and operated under as close to the design conditions specified in 3.7.1 as practicable and in accordance with the conditions specified in 4.6.8.2 through 4.6.8.2.2 for not less than two 240-hour production periods of actually delivering liquid oxygen at full production capacity into a . receiving container or tank maintained at approximately atmospheric pressure and in which the liquid level is at not less than 8 feet above the ground, floor, or surface on which the generating plant is operated. Operation time during which liquid oxygen is not actually being delivered into the container or tank at or in excess of the production rate and purity specified in 3.7.1 and 3.7.4.1 shall not be counted toward completion of the 240-hour production period. The time to achieve rated production at the beginning of the production period and after each shutdown shall not exceed the times allowed by 3.7.2 and 3.7.3, and the generating plant shall complete the production period without necessity for shutdown, thaw, or production break other than as permitted by 4.6.8.2 and 4.6.8.2.1, or the equipment shall be considered to have failed the test.

4.6.8.3.1 Immediately upon completion of a 240-hour liquid oxygen production period, the generating plant shall be thawed, serviced, and prepared for another 240-hour production period as specified in 4.6.8.1. The time required to accomplish the preparation for another 240-hour production period shall be reported, and shall not exceed the limit specified in 3.6.4 or the generating plant shall be considered to have failed the test.

4.6.8.4 Liquid nitrogen production phase. - The generating plant shall be adjusted for production of liquid nitrogen, started, and operated rider as close to the design conditions specified in 3.7.1 as practicable and in accordance with the conditions specified in 4.6.8.2 through 4.6.2.2.2 for not less than two 240-hour production periods of actually delivering liquid nitrogen at full production capacity into a receiving container or tank maintained at approximately atmospheric pressure and in which the liquid level is at not less than 8 feet above the ground, floor, or surface on which the generating plant is operated. Operation time during which liquid nitrogen is not actually being delivered into the container or tank at or in excess of the production rate and purity specified in 3.7.1 and 3.7.4.1 shall not be counted toward completion of the 240-hour production period. The time to achieve rated production at the beginning of the production period and after each shutdown shall not exceed the times allowed by 3.7.2 and 3.7.3, and the generating shall complete the production period without necessity four shutdown, thaw or production break other than permitted by 4.6.8.2 and 4.6.8.2.1, or the equipment shall be considered to nave failed the test.

4.6.8.4.1 Immediately upon completion of the 240-hour liquid nitrogen production period, the generating plane shall be thawed, serviced, and prepared for another 240-hour production period as specified in 4.6.8.1. The time required to accomplish the preparation for another 240-hour production period shall be reported, and shall not exceed the limit specified in 3.6.4 or the generating plant shall be considered to have failed the test.

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4.6.9 Night operation test. - The generating plant shall be operated in darkness sufficiently to demonstrate the adequacy of the lighting system specified in 3.11.10 and of the blackout lighting system specified in 3.11.9 for permitting safe and effective utilization under field conditions when no lighting other than that furnished as part of the generating plant is available. This evaluation shall be accomplished by operating the generating plant for not less than one complete 24C-hour phase of the endurance test specified in 4.6.8 using only the lights furnished on the generating plant and with the generating plant set up in a location where no outside illumination other than the natural sky light from the sun, moon, and stars will be received. The generating plant shall be operated throughout all hours of darkness without illumination other than the blackout lights for not less than five nights of the period. All instances where the illumination is inadequate for safe and effective operation of the equipment shall be reported in detail. If the illumination supplied by the lighting systems, both the regular and blackout systems, provided on the generating plant is inadequate in any manner for safe and effective operation of the equipment at night, the generating plant shall be considered to have failed the test.

4.6.10 Radiated electromagnetic interference test. - The generating plant shall be tested for radiated electromagnetic interference in accordance with the provisions specified in MIL-STD-462 for RE05, class IIIC, equipment while it is operating to deliver liquid product at maximum capacity. The radiated interference shall not exceed the limits established by MIL-STD-461 for RE05, class IIIC, items, or the generating plant shall be considered to have failed the test.

4.6.11 Accoustic noise level testing. - The generating plant shall be tested for accoustic noise level while it is operating to deliver liquid product at maximum capacity. The acoustic noise levels measured as specified in 4.6.11.1 shall not exceed the levels specified in 3.7.6, or the generating plant shall be considered to have failed the test.

4.6.11.1 Method of test. - The generating plant shall be set up in a level, open, space or field rather than in an enclosed space or near a wall. The generating plant shall then be started, and operated to produce liquid at maximum capacity. While so operating and producing liquid, the generating plant shall be tested for audio noise level with the measuring instrument sound level pickup device positioned at:

(a) 5-1/2 ft. ± 1/2 ft. above ground level at all control, adjustment, and operating positions
(b) 5-1/2 ft. ± 1/2 ft. above-ground level around the periphery of the generating-plant at 10 ft. ± 1 ft. from the equipment housing.

4.6.11.2 Sound level measuring instrumentation. - The audio sound level measurements shall be taken using a sound meter in conjunction with an octave band analyzer, The instruments used shall meet the requirements specified in S1.4 and Z24.10, except that the octave band analyzer shall employ the preferred frequencies specified in S1.6. The flat frequency response (C! weighting network) of the sound level meter shall be used in making all measurements. The instrumentation shall have a frequency response range of at least from 45 to 11,300 Hz with the response known within ± 3dB. An acoustical calibrator shall be used for an end-to-end (from the microphone to the meter) system calibration to provide the information needed for correcting the measured levels. The sound pressure-level (in dB (0,0002 dynes/Cm^2)) produced by the calibrator when used in-conjunction with the particular sound level pickup employed in the system-shall be known to within ± 0.5dB.

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4.6.12 Environmental testing. - The generating plant shall be subjected to the following tests conducted in accordance with the specified methods of MIL-STD-810, or as otherwise required by the individual test instructions.

4.6.12.1 Low temperatures tests. -

4.6.12.1.1 Low temperature exposure test. - The generating plant shall be subjected to a low temperature exposure in accordance with Method 502, procedure I of MIL-STD-810, except that the exposure temperature shall be -65° F instead of the -80° F specified in procedure I.

4.6.12.1.2 Low temperature start and operation test. - Upon completion of the low temperature exposure, the test chamber temperature shall be raised to -30 ± 5° F and the generating plant permitted to temperature stabilize. After the generating plant has stabilized at -30° F, all operating components, controls, valves, and accessories shall be checked for free operation; all cables, hoses, and attachments shall be flexed or exercised as in normal use; and each door or hinged access opening shall be opened, closed, and latched not less than 10 times. After these operations and while the equipment is still at the specified temperature, the generating plant shall be started, without preheating or other external aids, and operated at least until liquid production begins. The generating plant shall start and operate without heating or other external aids, and provide weather shelter and comfort heating for the operators, or shall be considered to have failed the test.

4.6.12.1.3 Inspection of low temperature test effects. - Following the low temperature testing, the generating plant shall be permitted to warm to normal ambient temperature. It shall then be inspected for any damage or deterioration resulting from the low temperature, exposure, starting, and operation. Any damage or deterioration resulting from the low temperature testing that could degrade the equipment life, usefulness, reliability, maintainability, or resistance to environmental or other damage shall be cause for rejection.

4.6.12.2 High temperature tests. -

4.6.12.2.1 High temperature exposure test. - The generating plant shall be subjected to high temperature exposure in accordance with Method 501, procedure I, except that the exposure temperature shall be 165° F instead of the 160°F specified in procedure I.

4.6.12.2.2 High temperature start and operation test. - Upon completion of the testing and inspection at 165° F, the chamber temperature shall be reduced to and maintained at 125 ± 5° F for not less than 24 hours. The relative humidity shall be raised to not less than 75 Percent during this period. Upon completion of the exposure and while still at the specified temperature and relative humidity, the generating plant shall be started and operated until liquid oxygen has been delivered for a sufficient length of time to determine its production capacity, but in no case less than a minimum of 4 hours of production at maximum output. The generating plant shall start and operate as necessary for compliance with 3.6.2 and 3.7.1, or shall be considered to have failed the testing.

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4.6.12.2.3 Inspection of high temperature test effects. - Following the high temperature testing, the generating plant shall be permitted to cool to normal ambient temperature, It shall then be inspected for any damage or deterioration resulting from the high temperature exposure, starting, and operation, Any damage or deterioration resulting from the high temperature testing that could degrade equipment life, usefulness, reliability, maintainability, or resistance to environmental or bther damage shall be cause for rejection.

4.6.12.3 Humidity test. - The complete generating plant shall be subjected to a humidity test in accordance with Method 507, procedure I of MIL-STD-810. All housing assembly doors and hinged access openings shall be open throughout this test,

4.6.12.4 Fungus test. - Equipment samples adequately representing applicable portions of the generating plant that might be damaged by exposure to fungus attack shall be subjected to fungus in accordance with Method 508, procedure I.

4.6.12.5 Salt-fog test. - The generating plant shall be subjected to a salt-fog test in accordance with Method 509, procedure 1,

4.6.12.6 Rain test. - The complete generating plant shall be subjected to a rain test-dance with Method 506, procedure I.

4.6.12.7 Sand and dust test. - The generating plaint while operating, shall be subjected to a sand and dust test in accordance with Method 510, procedure I.

4.6.12.8 Wind test. - It shall be demonstrated, by testing or calculations, or both, that 'sting plant will withstand without upset or permanent deformation of parts, the 70 mph wind velocity specified in 3.6.3.9 both when prepared for shipment and when set up and operating.

4.6.12.9 Snowload test. - The generating plant shall be subjected to simulated snowload of 40 psf both when closed for transport and when configured for operation in cold climates. Damage to or permanent deformation of the housing or of other parts shall be cause for considering the generating plant to have failed the test.

4.6.13 Servicing and maintenance tests. - All normal servicing and preventive maintenance operations specified in the generating plant operating, servicing, and maintenance instruction handbooks prepared by the contractor shall be performed to determine their adequacy, ease of accomplishment, and the accessibility of parts and assemblies for performance of same. Insofar as practical, these tests shall be performed as part of the normal preventive maintenance, servicing, and inspection operations performed in accomplishing the other testing specified herein. The servicing and maintenance performed shall include removal and replacement of the engine package. Interferences or obstructions to servicing or preventive maintenance and repair shall be reported in detail.

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4.6.13.1 Special tool utilization. - All generating plant special tools and field equipment shall be used to the maximum practicable extent during the testing specified herein. Utilization of the special tools and field equipment shall be sufficient to determine their usefulness. Instances where the special tools furnished are inadequate shall be reported in detail.

4.6.14 Reliability demonstration. - Satisfactory completion of all tests specified herein with no failure that could prevent satisfactory or safe performance under the required conditions shall be considered as demonstrating acceptable compliance with the quantitative reliability requirements of this specification. Any test resulting in a failure shall be thoroughly documented. Apparent failures that are caused by some unintended factor external to the equipment (for example, test equipment failure) need not be counted as reliability failures if this causes can be substantiated to procuring activity satisfaction. Acceptance for reliability purposes shall not be made until, to the satisfaction of the procuring activity, each failure has been analyzed, the cause has been determined, design or other modifications have been made to preclude repetition of the failure or related failures, the contractor has conducted successful retest of the failed equipment with the modifications included, and the contractor has taker. action to reflect the modifications in equipment to be delivered. Any retest shall include all tests, in addition to a particular failed test, that could be influenced by the modifications,

4.6.15 Maintainability demonstration. - Compliance with qualitative maintainability requirements shall be demonstrated by inspection and examination of the generating plant. Compliance with quantitative maintenance downtime requirements shall be demonstrated by task performances as specified below. The actual maintenance downtimes, the tools and equipment required, and any obstructions or conditions hindering task performance or presenting a potential hazard to personnel or to parts shall be observed and reported. Signs of excessive wear or other indications of potential failure shall be reported as to when and where observed, and the information supplied to the design activity for. study and for recommendations concerning appropriate corrections to be reflected in items to be delivered, Such recommendations shall be subject to procuring activity approval before being implemented, or successful repeat of other tests specified herein may be required where the test result could be influenced by the corrections. It shall be demonstrated that the generating plant is in satisfactory operating condition after the maintenance tasks have been completed. Maintenance task time data from other maintenance and teardown inspections done in connection with other tests specified herein may be applied to the maintainability demonstration if adequate observations are made and reported.

4.6.15.1 Corrective maintenance. - Compliance with the corrective maintenance downtime requirements shall be established by a maintenance demonstration procedure, maintenance task selection, and maintenance task performances in accordance with MIL-STD-471 including appendices A and B, Method 1, test plan paragraph A1 plus B1. The tasks selected and done shall be related to the recommended or to the stated and assumed maintenance concept.

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4.6.15.2 Preventive maintenance. - Compliance with the preventive maintenance downtime requirements shall be demonstrated in accordance with MIL-STD-471, appendix Method 6. Each task need not be done more than once, but the resulting times shall be weighed in calculations to account for the expected frequency of each task in the reference period selected. For maximum preventive mainter.ante down the demonstration, less than 0.1 (10 percent) of the task times observed shall require greater downtime than the quantitative requirement, Preventive maintenance tasks shall include servicing and any calibrations required.

4.6.16 Mechanical check. - Upon completion of the other testing specified herein, a critical inspection shall be made of all components to determine their operability and any damage or undue wear incurred during the testing. Teardown and parts measurement shall be performed only in those cases where service life is in question. Where teardown and parts measurement are performed, wear or distortion that exceeds limits permitted by the manufacturer for new parts shall be cause for considering the part of parts affected as having failed to satisfactorily complete the test.

4.6.16.1 Magnetic particle inspection. - Ferrous or other magnetic material parts subject to high stresses in operation and which are suspected of having detects shall also be subjected to magnetic particle inspection (magnaflux) in accordance with MIL-I-6868. The parts shall exhibit no indication of damage attributable to the testing, or they shall be considered as having failed the testing.

4.6.16.2 Fluorescent penetrant inspection. - Nonmagnetic castings, forgings, and other parts suspected of defects shall be subjected to inspection with fluorescent penetrant (black light) in accordance with MIL-I-6866. indications or evidence of defects or damage attributable to the testing shall be cause for considering a part as having-failed the testing.

4.7 Inspection of preparation for delivery. - Preservation, packaging, packing, and marking shall be inspected to determine compliance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 General instructions. -

5.1.1 Cleaning. - The generating plant snail be thoroughly cleaned to remove excess and spilled lubrication materials, loose or chipped paint, spilled chemicals, loose bolts, nuts, screws, and other foreign materials. All cleaning solvents and similar materials shall be thoroughly removed from the generating plant components and accessories prior to delivery.

5.1.1.1 Parts contacting product. - The generating plant parts, fittings, components, and assemblies that contact oxygen or nitrogen products shall be thoroughly cleaned and dried by one of, or by a combination of the following methods:

Method A. Vapor degreasing process using stabilized trichloro-ethylene conforming to O-T-634 in a standard commercial vapor degreaser. Operation of-the degreaser shall be in accordance with the manufacturer's recommendations, All solvents shall be removed after the degreasing operation by purging with hot, oil-free air or nitrogen, or by vacuum evacuation.

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Method B. Solvent degreasing by thoroughly washing with solvent conforming to MIL-T-27602. All solvents shall be removed after the degreasing operation by purging with hot, oil-free air or nitrogen, or by vacuum evacuation. Precautions shall be taken to insure the chlorinated solvent does not contact plastics or other incompatible materials.

Method C. Alkali detergent decreasing by thoroughly flushing with hot alkaline cleaning compound conga . . l ' conforming to P-C-436 until the components and surfaces are free from oil, grease, and other combustible materials, Following alkaline cleaning, all surfac es, internal and external, as applicable, shall be thoroughly rinsed with clean, hot water, and dried by purging with hot> oil-free air or nitrogen or by baking at a temperature of 250° to 300° F until all water is removed,

Method D. Ultrasonic cleaning with water base detergents or organic solvents. Care must be taken to insure that the solution used will not attack or erode the parts being cleared. Water base detergent solutions shall be used for cleaning simple unchannel parts. Detergent shall conform to type I of MIL-D-16791. Trichloroethylene shall be used for cleaning complex parts. Trichloroethylene shall conform MIL-T-27602.

5.1.2 Repainting. - Generating plant surfaces and parts on which the paint or other protective coating is damaged or defective shall be thoroughly cleaned and repainted, or refinished as specified under 3.1.7.

5.1.3 Lubrication. - All generating plant machined bearing surfaces that are normally Lubricated in operation shall be thoroughly lubricated with the recommended lubricants so as to be ready for immediate operation,

5.1.4 Disassembly and matchmarking. -

5.1.4.1 Disessembly. - Unless otherwise specified, the extent of disassembly shall be confined to the removal of parts that would otherwise increase cubage or would be vulnerable to damage, and shall be in accordance with the recommendations of the manufacturer, subject to the approval of the procuring activity. All bolts, nuts, pins, washers and other fastening or retaining devices removed during disassembly shall be replaced in one of the mating pares and secured in a manner that will prevent loss.

5.1.4.2 Mate.hmarking. - Matchmarking information and identification shall be printed on water-resistant cloth shipping tags conforming to type A, size 1 of UU-T-81, securely attached to the removed parts and mating parts on the basic unit in such a manner that the items will not be damaged, Tags and printing thereon shall be waterproofed.

5.1.5 Component draining. - The column reboilers, liquid receivers, traps, and other components in which water or cleaning solvents may accumulate shall be thoroughly drained and purged, Following this draining and purging, the drain valves shall be closed and the drain openings sealed with pressure-sensitive tape conforming to PPP-T-60.

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5.1,5.1 Warning tag. - A type A tag conforming to UU-T-81, with tag and printing waterproofed, stating that the liquid drains have been closed and sealed with pressure-sensitive tape, shall be securely attached on the generating plant main control panel in a conspicuous Location.

5.1.6 Preservatives. - Preservatives or lubricants shall not be applied to any part of the equipment that will come into contact with oxygen or nitrogen product.

5.1.7 Securing for shipment. - Small items and packages not fastened to major components shall be secured to or be packaged with major components to insure against accidental loss or damage caused by movement during shipping.

5.2 Preservation and packaging. - Preservation and packaging shall be level A or level C, as specified (see 6.2),

5.2.1 Level A. - The generating plant shall be preserved and packaged for extended storage or overseas shipment as directed by the procuring activity.

5.2.2 Level C. - The generating plant shall be preserved and packaged in accordance with the manufacturers commercial practice for domestic delivery and short-term storage,

5.3 Packing. - Packing shall be level A, level B, or level C, as specified (see 6.2).

5.3.1 Level A. - The generating plant shall be packed for shipment and storage as directed by the procuring activity.

5.3.2 Level B. - The generating plant shall be packed for shipment and storage as directed by the procuring activity.

5.3.3 Level C. - The generating plant shall be packed for shipment in accordance with the manufacturer's commercial practice in a manner that will insure safe transportation, at the lowest rate, to the point of delivery. Shipping packages and containers shall meet Uniform Freight Classification Rules or regulations of other common carriers as applicable to the mode of transportation. Items that require overpacking for acceptance by the carrier shall be packed in exterior-type shipping containers.

5.4 Physical Protection. - Cushioning, blocking; bracing, and bolting as required for compliance with the requirements specified herein shall be in accordance with MIL-STD-1186, except that for domestic shipments, water-proofing requirements for cushioning materials and containers shall be waived. The drop tests of MIL-STD-1186 shall be waived when preservation packaging, and packing of the item are for immediate use or when the drop tests of MIL-P-116 are applicable.

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5.5 Shipment marking. - Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Generating Plant,

Oxygen-Nitrogen,

Trailer Mounted,

A/M26U-1

6. NOTES

6.1 Intended use. - The A/M26U-1 generating plant covered by this specification is intended for use in producing high-purity liquid oxygen and liquid nitrogen for support of advance tactical airbase requirements.

6.2 Ordering data. - Proacurement documents should specify the following:

(a) Title, number, and date of this specification
(b) When the generating plant is to be produced in accordance with the Air Force special requirements
(c) When the generating, plant is to be produced in accordance with the Navy Department special requirements
(d) When a complete set of all special tools, other than common handtools and those normally available in motor vehicle repair shops, is to be provided
(e) Location and conditions for preproduction testing
(f) When instruction manuals covering operation, servicing, and maintenance of the generating plant, or other technical publications, (including identification, quantities, and disposition) are to be provided with each unit
(g) Require level of preservation, packaging, and packing (sic Section 5)
(h) The portable heater for which the generating plant winterization provisions shall provide (see 3.5.10)
(i) The connections or connectors to be provided for use of an auxiliary air source (see 3.8.2.6)
(j) The fuel tank fill connection to be supplied (see 3.10)
(k) The liquid product transfer hoses and servicing end corrections to be supplied (see 3.13.2)
(l) The type fire extinguishers and brackets to be supplied on the generating plant (see 3.13.3)
(m) The intercommunication system to be provided on the generating plant (see 3.14)
(n) The generating plant colors (see 3.18)
(o) The color to be used for stenciling markings on the generating plant (see 3.19.2.1)
(P) Service name and registration numbers (see 3.19)

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6.3 Definitions. For the purpose of this specification, the following definitions shall apply:

6.3.1 Ambient conditions. - Ambient conditions are the dry-bulb temperature, wet-bulb temperature, and the relative humidity of the atmospheric air surrounding, and in the vicinity of, the generating plant, but unaffected by any heat, cold, or moisture emanating from the generating plant itself.

6.3.2 Production period. - A production period is defined as a period of time during which oxygen and nitrogen product is actually being delivered at full capacity into receiving containers.

6.3.3 Pressures. - Unless specified as absolute or denoted by the symbol psia, all pressures referred to herein will be interpreted as pounds per square inch (psi) gage.

6.3.4 Valve cycle of operation. - A valve cycle of operation is defined as opening the valve from sealing a specified differential pressure across the valve to the not less than 3/4 open position or the full open position (as specified), and then reclosing the valve with a specified torque,

6.3.5 Fibrous particle, - A fibrous particle is defined as a long, slender particle whose maximum cross sectional dimensions is 40 microns.

6.3,6 Improved road. - An improved road is a smooth, hard-surfaced road - such as a concrete or asphalt paved highway (see 4.6.6.1.1).

6.3.7 Unimproved road. - An unimproved road is an unsurfaced road of dir gravel, or similar construction.

Custodians:
Navy - AS
Air Force - 11

Preparing activity:
Navy - AS

Project No. 3655-N030

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MIL-G-8968(ASG) GENERATING PLANT, OXYGEN-NITROGEN, TRAILER MOUNTED A/M 26U-1

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

( ) Direct Government Contract ( ) Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE?

( ) YES ( ) NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity) DATE

DD Form 1426

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